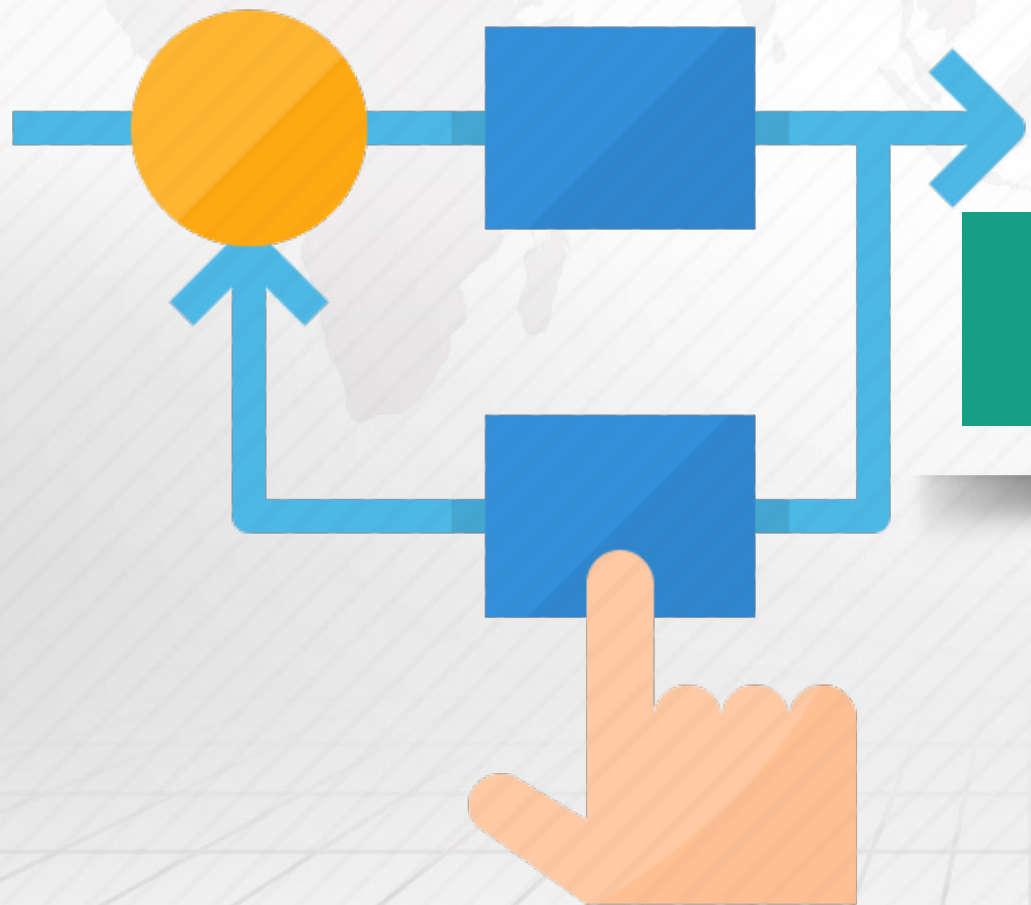




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JIANGSU UNIVERSITY OF SCIENCE AND TECHNOLOGY

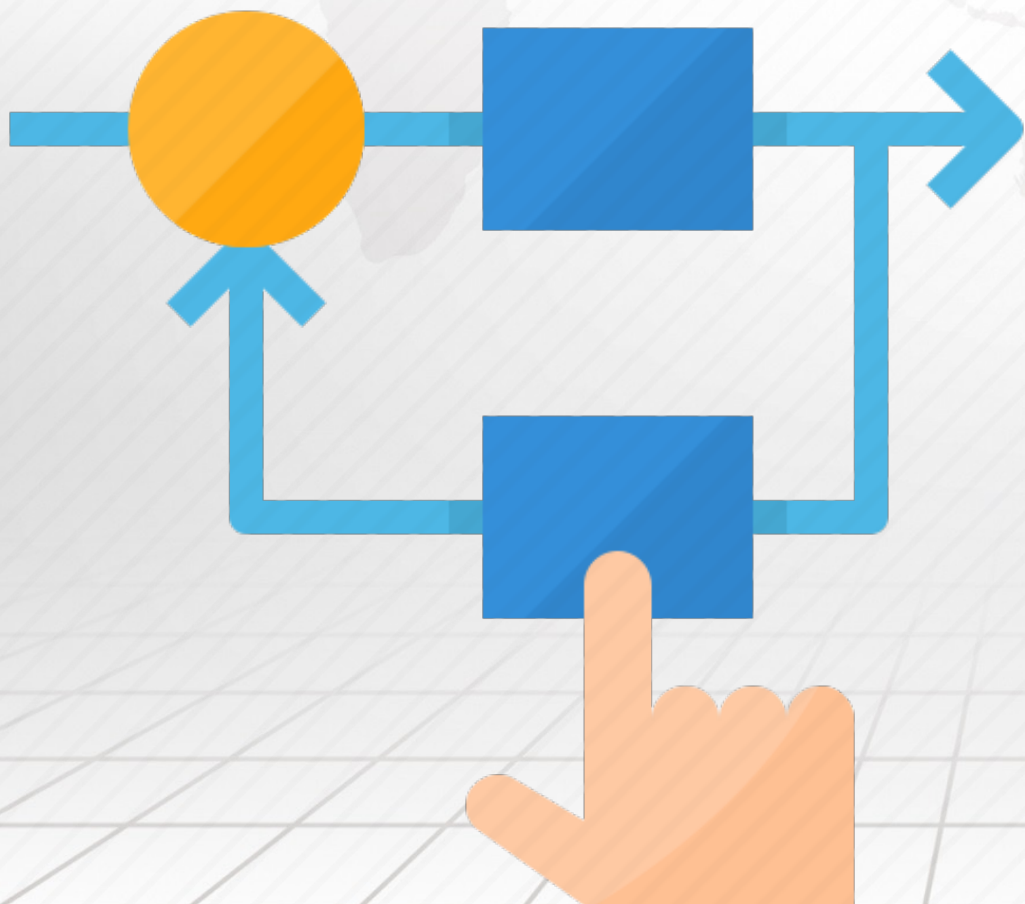


自动控制原理

第一章	第二章	第三章	第四章	第五章	第六章
基本概念	数学模型	时域分析法	根轨迹法	频域分析法	校正方法

10. 时域分析综合

学习目标



01

能够设定实际对象性能指标

02

能够综合运用时域分析手段

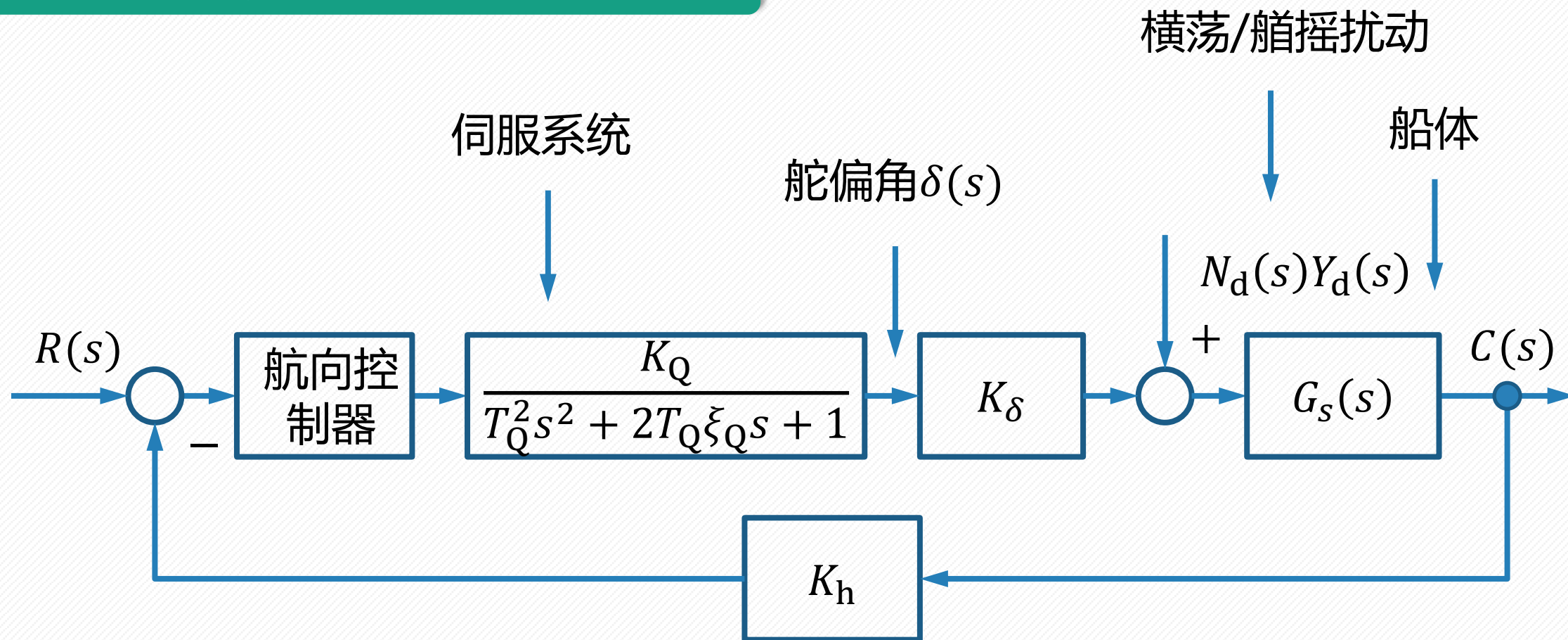
难

能够合理选择校正装置

重

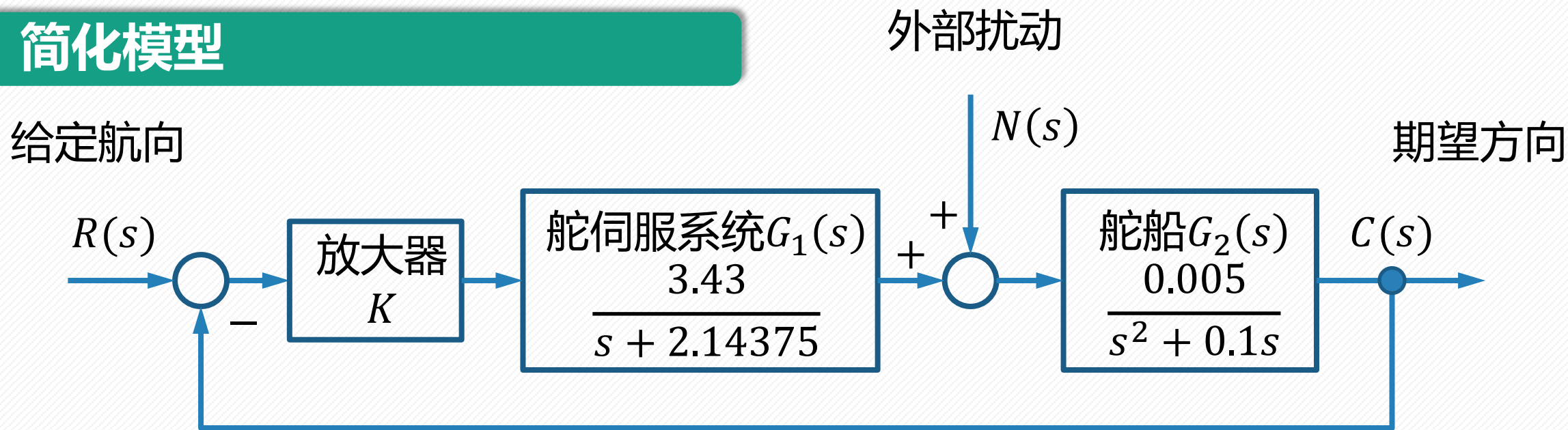
能够运用计算机辅助验证校正效果

航向系统模型



$$G_s(s) = \frac{1}{s(T_\psi s + 1)} \text{ 或 } G_s(s) = \frac{\tau_3 s + 1}{s(\tau_1 s + 1)(\tau_2 s + 1)}$$

简化模型

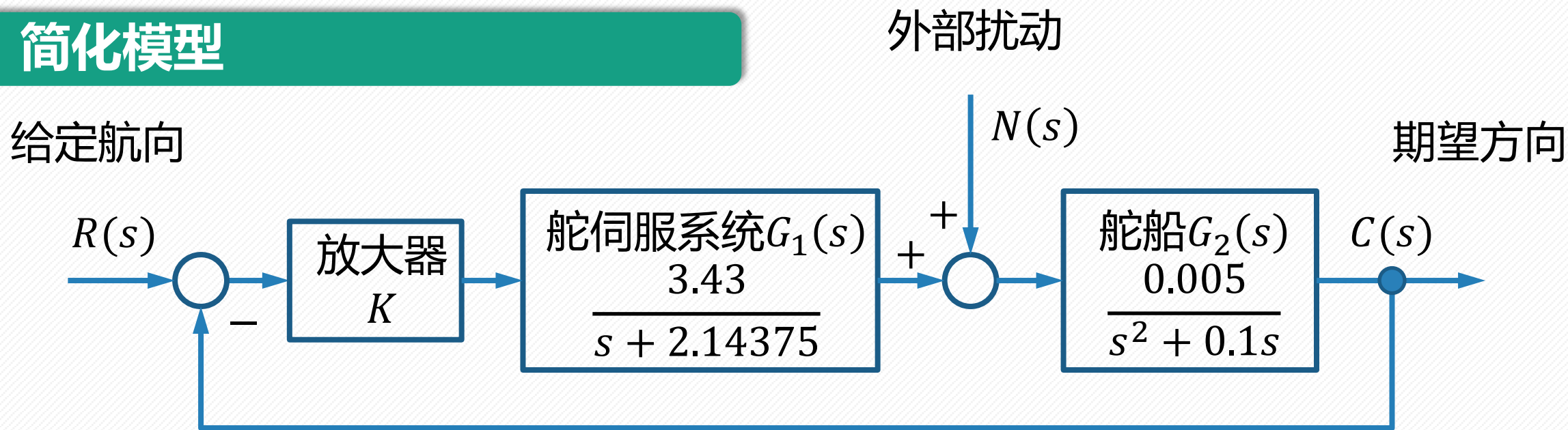


1 输入信号有哪些形式？代表什么物理含义？如何减小稳态误差？

2 扰动信号有哪些形式？代表什么物理含义？对船体产生的影响要如何分析？

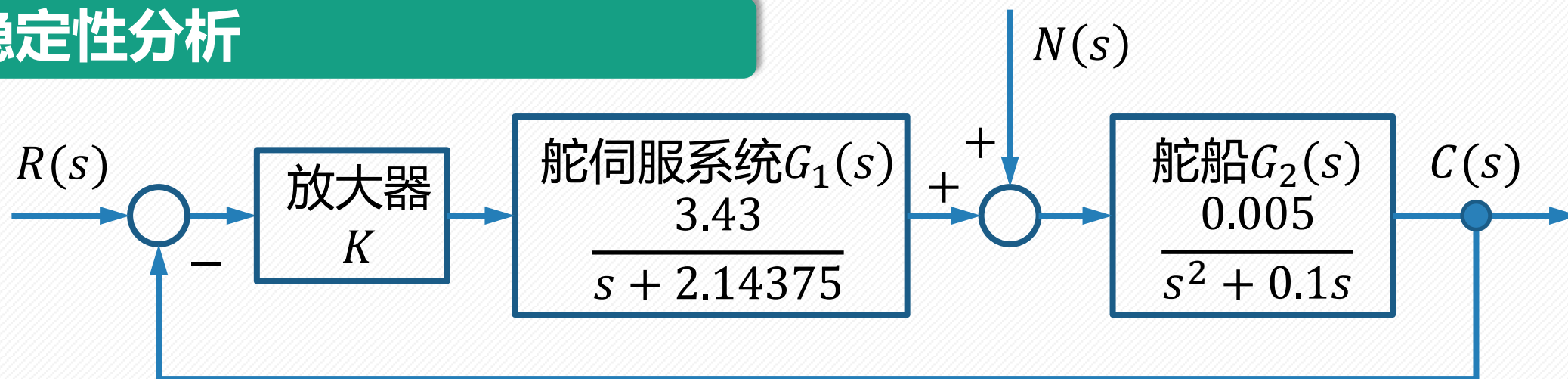
3 典型工况下输入和扰动分别使用什么信号模拟？

简化模型



- 1 分析参数 K 对闭环系统性能的影响，选择 K 使斜坡输入和阶跃扰动下的稳态误差较小。
- 2 分析单位阶跃扰动和正弦扰动下的开环输出响应。
- 3 分析单位阶跃输入和正弦扰动共同作用下的输出响应。

稳定性分析



闭环传函

$$\Phi(s) = \frac{C(s)}{R(s)} = \frac{K G_1(s) G_2(s)}{1 + K G_1(s) G_2(s)}$$

开环传函

$$K G_1(s) G_2(s) = \frac{0.01715K}{s(s + 0.1)(s + 2.14375)}$$

闭环特征方程

$$D(s) = 1 + K G_1(s) G_2(s) = 1 + \frac{0.01715K}{s(s + 0.1)(s + 2.14375)} = 0$$

无扰动情况分析

闭环特征方程

$$D(s) = 1 + K G_1(s) G_2(s) = 1 + \frac{0.01715K}{s(s + 0.1)(s + 2.14375)} = 0$$

$$D(s) = s^3 + 2.24375s^2 + 0.214375s + 0.01715K = 0$$

劳斯表

s^3	1	0.214375
s^2	2.24375	0.01715K
s^1	$0.214375 - 0.007643454K$	0
s^0	0.01715K	

稳定的充要条件

$$0 < K < 28.046875$$

设 $N(s) = 0$, $r(t) = Rt$

$$R(s) = \frac{R}{s^2}$$

斜坡作用下的稳态误差为

$$e_{ss} = \frac{R}{K_v}$$

无扰动情况分析

稳定的充要条件 $0 < K < 28.046875$

斜坡作用下的稳态误差为 $e_{ss} = \frac{R}{K_v}$

静态误差系数 $K_v = \lim_{s \rightarrow 0} s K G_1(s) G_2(s) = \lim_{s \rightarrow 0} s \frac{0.01715K}{s(s + 0.1)(s + 2.14375)} = 0.08K$

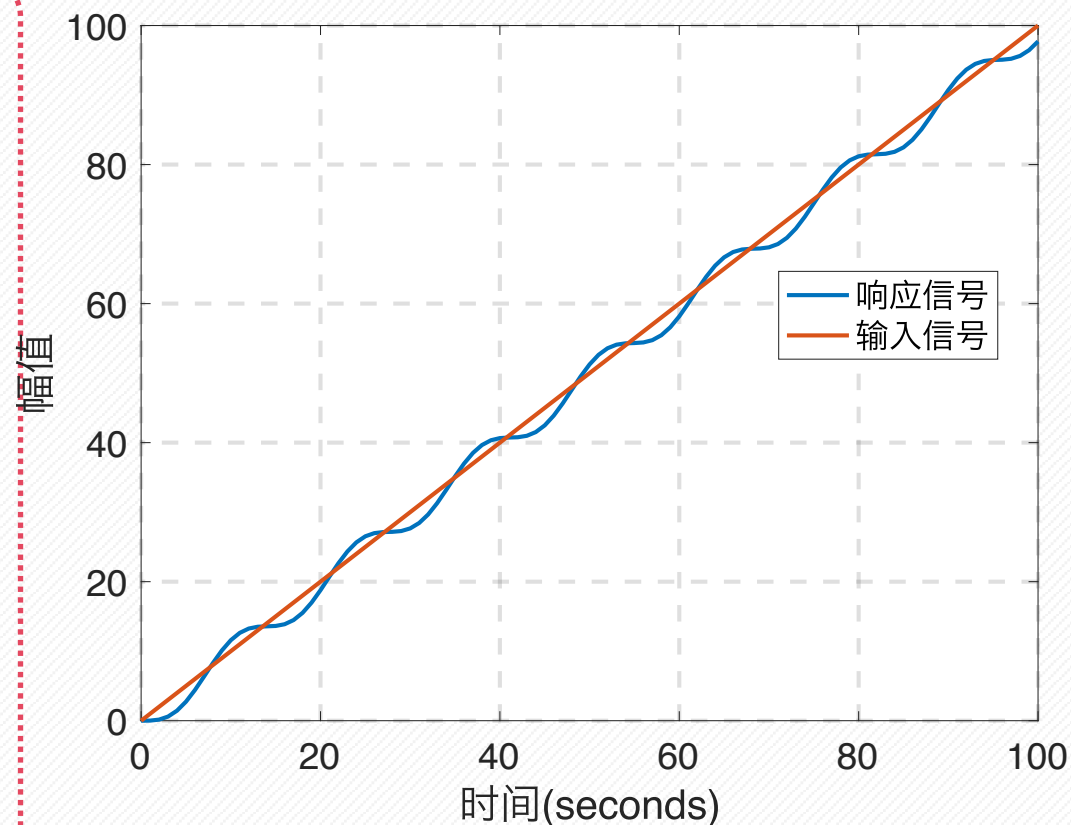
→ $e_{ss} = \frac{R}{0.08K}$

稳定的前提下，K越大，稳态误差越小

单位斜坡响应

代码

```
s = tf('s');
K = 28;
G1 = 3.43/(s+2.14375);
G2 = 0.005/(s*(s+0.1));
Phi = K*G1*G2/(1+K*G1*G2);
t = 0:100;
[y,t] = lsim(Phi,t,t);
plot(t,[y,t]);
xlabel('时间(seconds)')
ylabel('幅值');
legend({'响应信号','输入信号'}, 'Location', 'best');
grid on
ax = gca;
ax.GridLineStyle = "--";
ax.GridLineWidth = 2; % R2023a
```



串联校正

扰动作用下闭环传函

$$\Phi_n(s) = \frac{C(s)}{N(s)} = \frac{G_2(s)}{1 + K G_1(s) G_2(s)} = \frac{0.005(s + 2.14375)}{s(s + 0.1)(s + 2.14375) + 0.01715K}$$

阶跃扰动下稳态输出

$$C_n(\infty) = \lim_{s \rightarrow 0} s K \Phi_n(s) N(s) = \frac{0.625}{K}$$

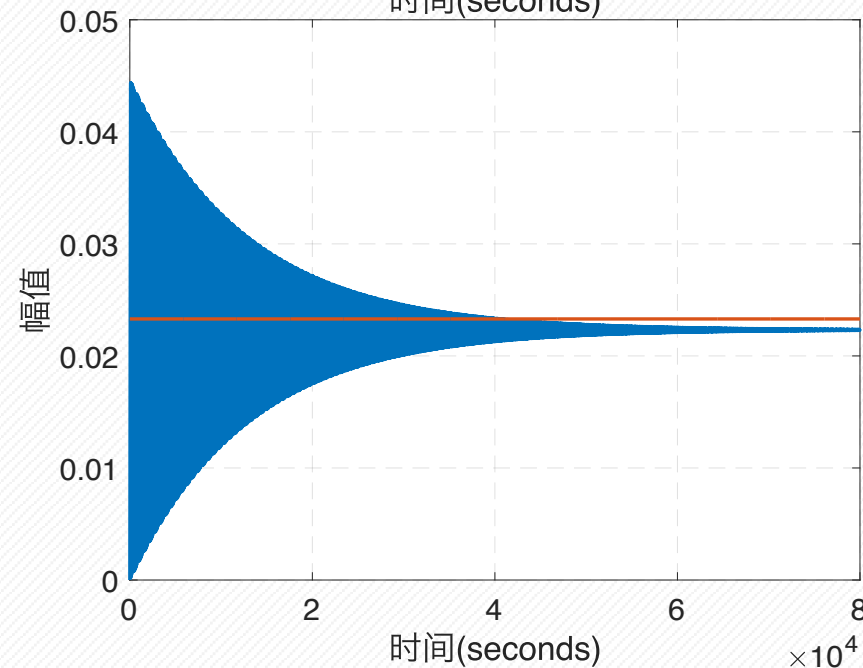
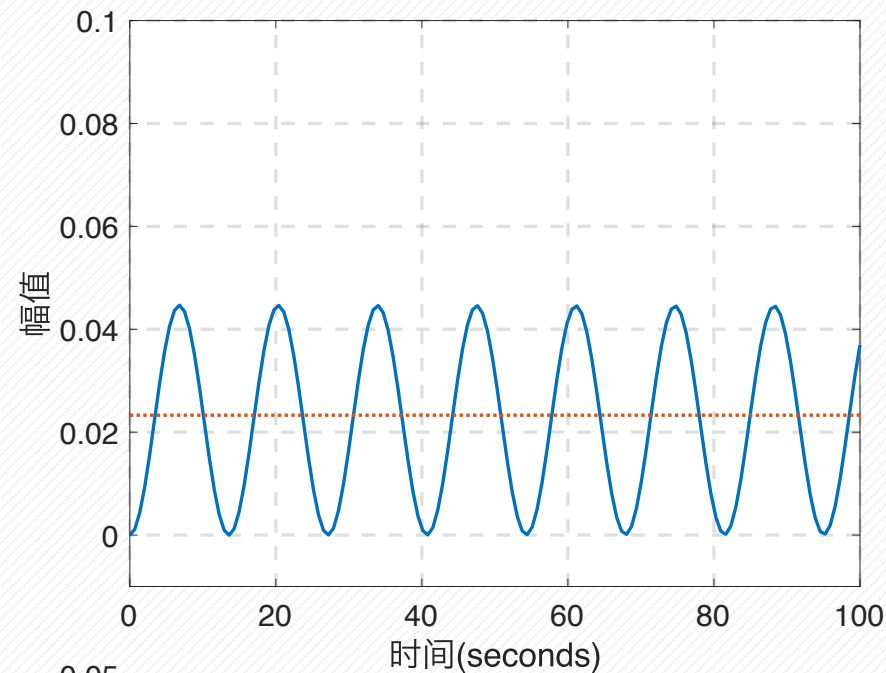
K 取28时

$$|C_n(\infty)| \leq 0.0233$$

单位阶跃扰动响应

代码

```
K = 28;  
Phi =  
0.005*(s+2.14375)/(s*(s+0.1)*(s+2.14375)+0.0171  
5*K);  
[y,t] = step(Phi,100);  
plot(t,y);  
hold on  
plot(t,ones(size(t))*0.0233,'LineStyle',':');  
xlabel('时间(seconds)')  
ylabel('幅值');  
grid on  
ax = gca;  
ax.GridLineStyle = "--";  
ax.GridLineWidth = 2; % R2023a  
ylim([-0.01,0.1]);
```



开环扰动分析

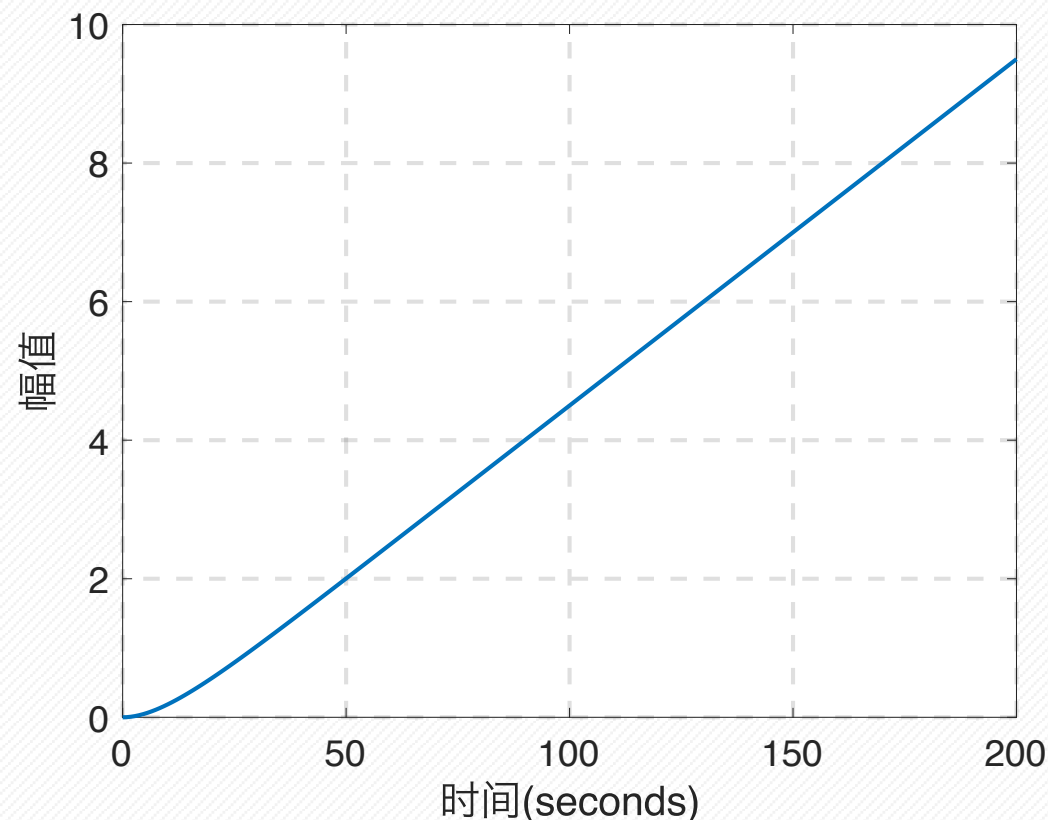
海浪有义波高2.5 ~ 5 m时，海浪主要能量集中频段频率为0.25 ~ 2.0 rad/s

significant wave height: 最高的1/3波浪的平均波高。人们在海上目测的波高很接近有义波高。

选择扰动因素信号频率为1.6 rad/s，即信号周期为10s

$$n(t) = \sin\left(\frac{2\pi}{10}t + \frac{\pi}{6}\right)$$

采用等效正弦模拟海浪



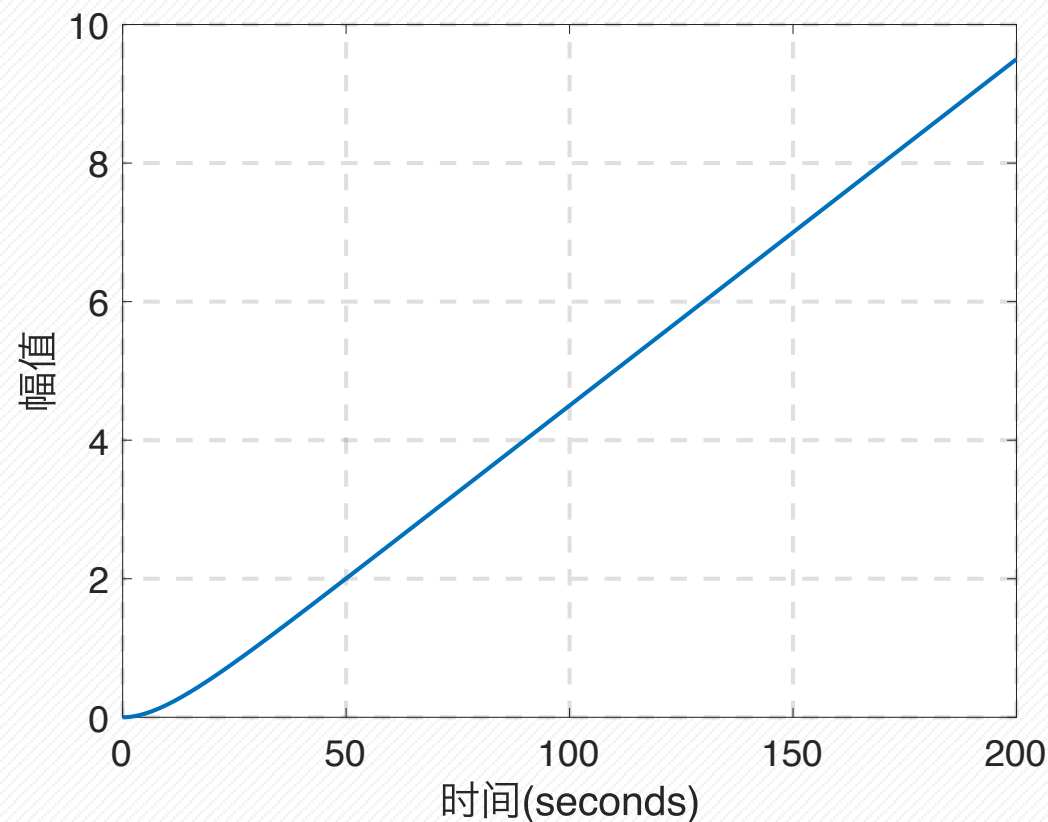
开环系统在单位阶跃扰动信号作用下的响应曲线

开环扰动分析

代码

```
G2 = 0.005/(s*(s+0.1));  
[y,t] = step(G2,200);  
plot(t,y);  
xlabel('时间(seconds)')  
ylabel('幅值');  
grid on  
ax = gca;  
ax.GridLineStyle = "--";  
ax.GridLineWidth = 2; % R2023a
```

采用等效正弦模拟海浪



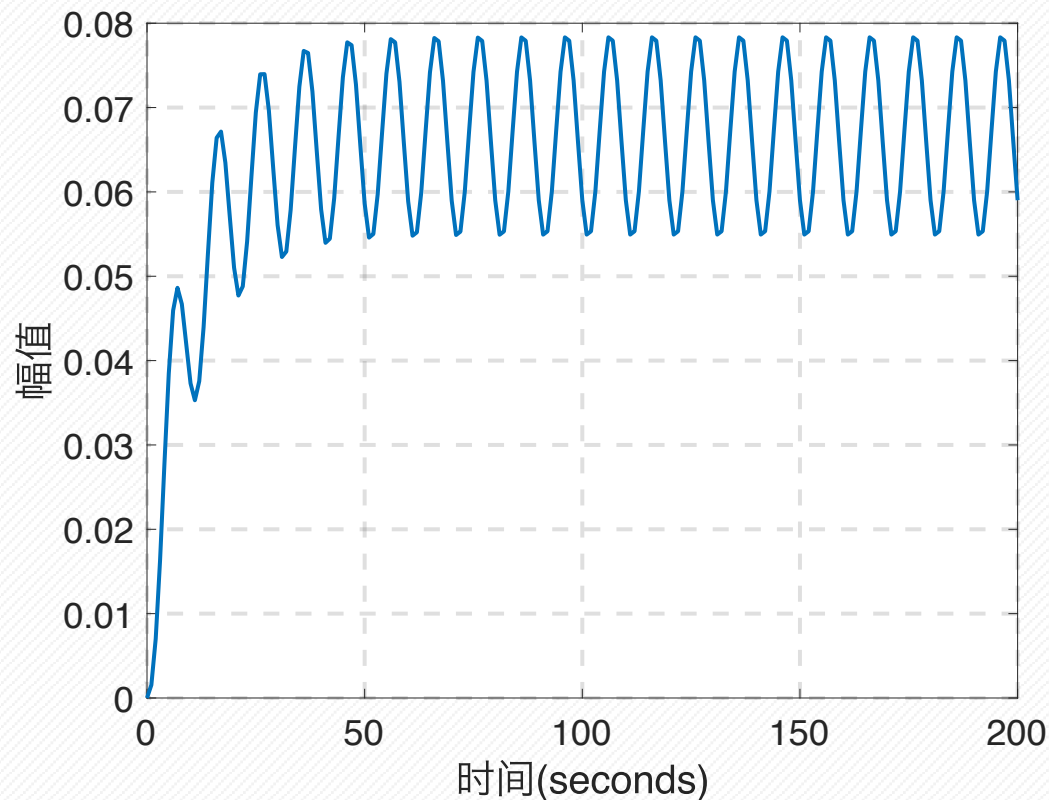
开环系统在单位阶跃扰动
信号作用下的响应曲线

开环扰动分析

代码

```
G2 = 0.005/(s*(s+0.1));  
t = 0:200;  
u = sin(2*pi/10.*t+pi/6);  
[y,t] = lsim(G2,u,t);  
plot(t,y);  
xlabel('时间(seconds)')  
ylabel('幅值');  
grid on  
ax = gca;  
ax.GridLineStyle = "--";  
ax.GridLineWidth = 2; % R2023a
```

采用等效正弦模拟海浪

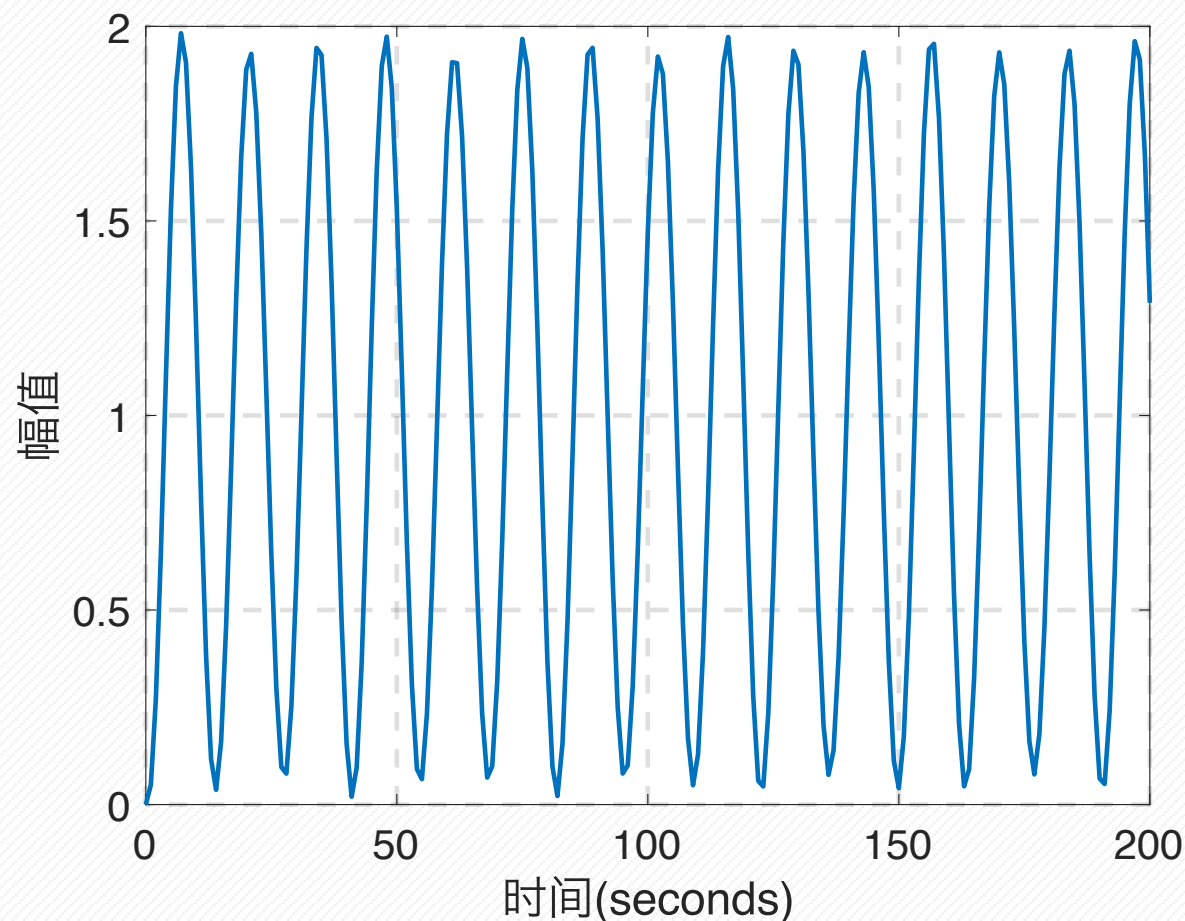


开环系统在等效正弦扰动
信号作用下的响应曲线

输入和扰动共同作用下的响应

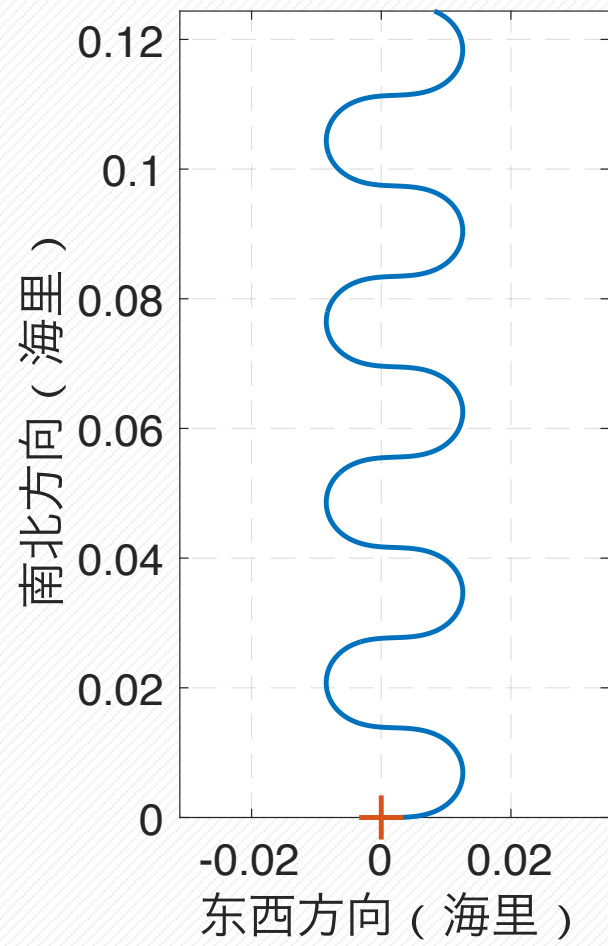
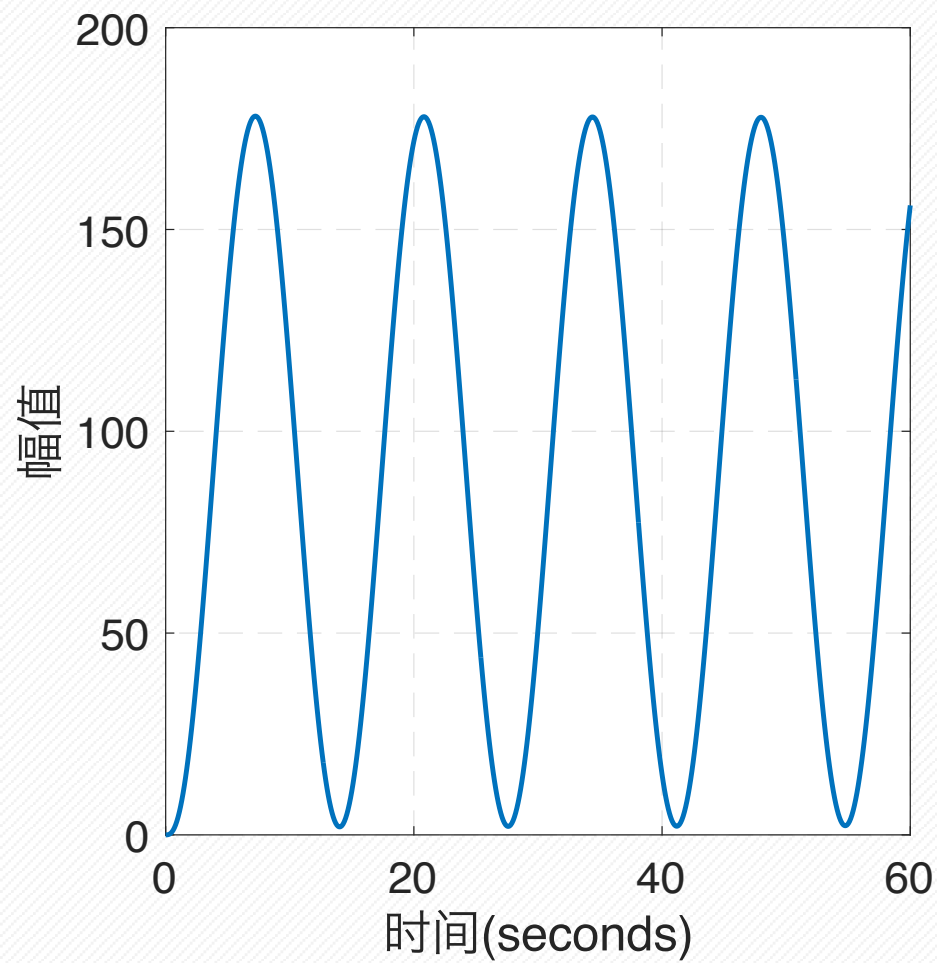
代码

```
K = 28;  
G1 = 3.43/(s+2.14375);  
G2 = 0.005/(s*(s+0.1));  
PhiR = K*G1*G2/(1+K*G1*G2);  
PhiN = G2/(1+K*G1*G2);  
t = 0:200;  
u = 1*ones(size(t));  
[y1,t] = lsim(PhiR,u,t);  
u = sin(2*pi/10.*t+pi/6);  
[y2,t] = lsim(PhiN,u,t);  
plot(t,y1+y2);  
xlabel('时间(seconds)')  
ylabel('幅值');  
grid on  
ax = gca;  
ax.GridLineStyle = "--";  
ax.GridLineWidth = 2; % R2023a
```

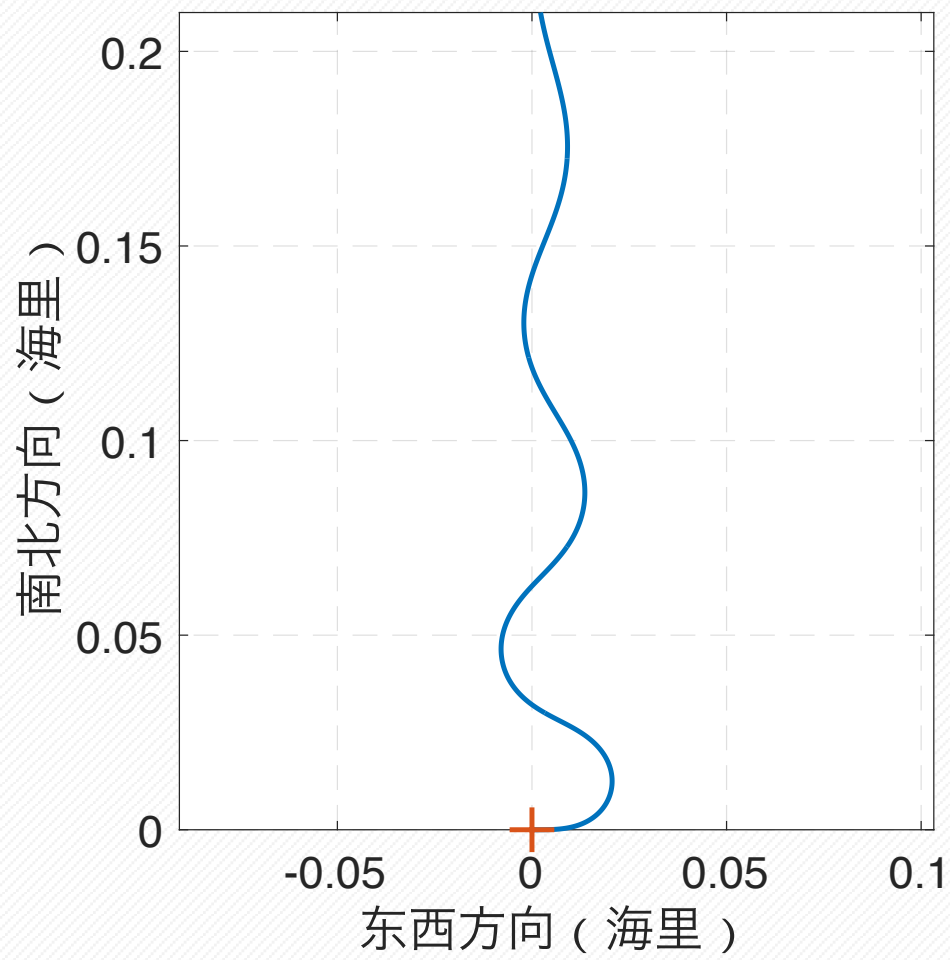
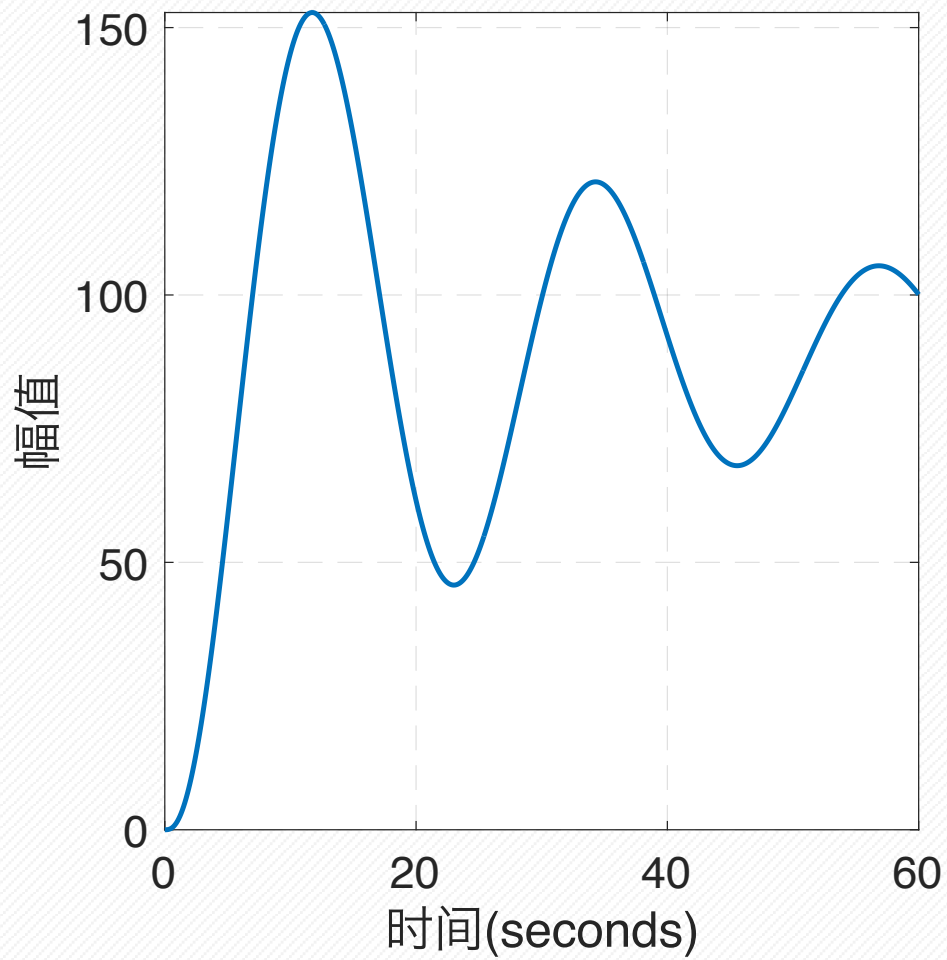


单位阶跃输入和等效正弦扰动信号
共同作用下的闭环系统响应曲线

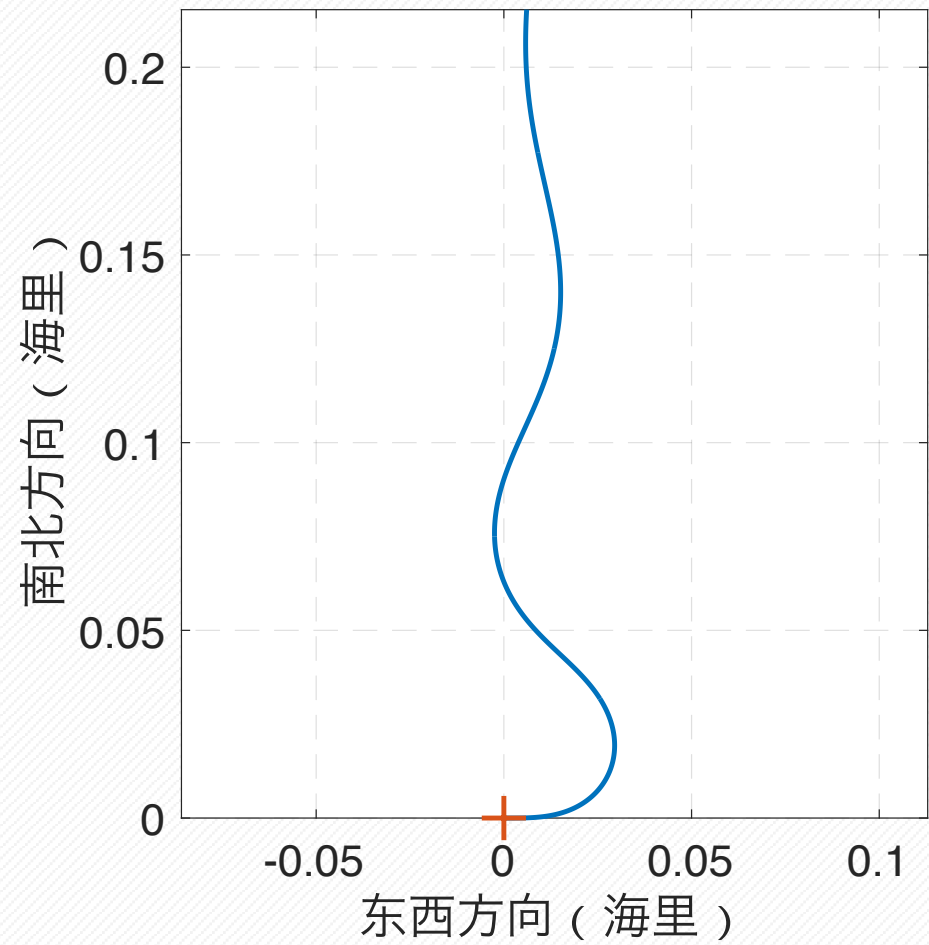
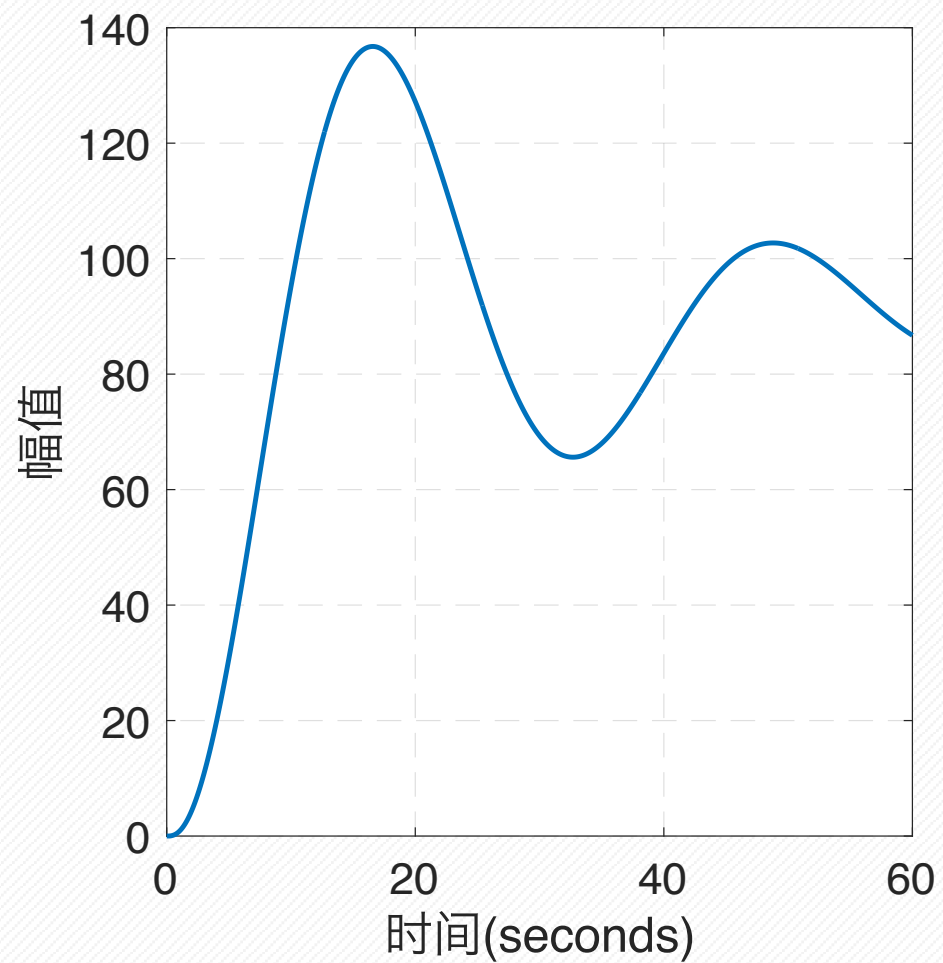
直角转向性能 ($K=28, V=15$ 节)



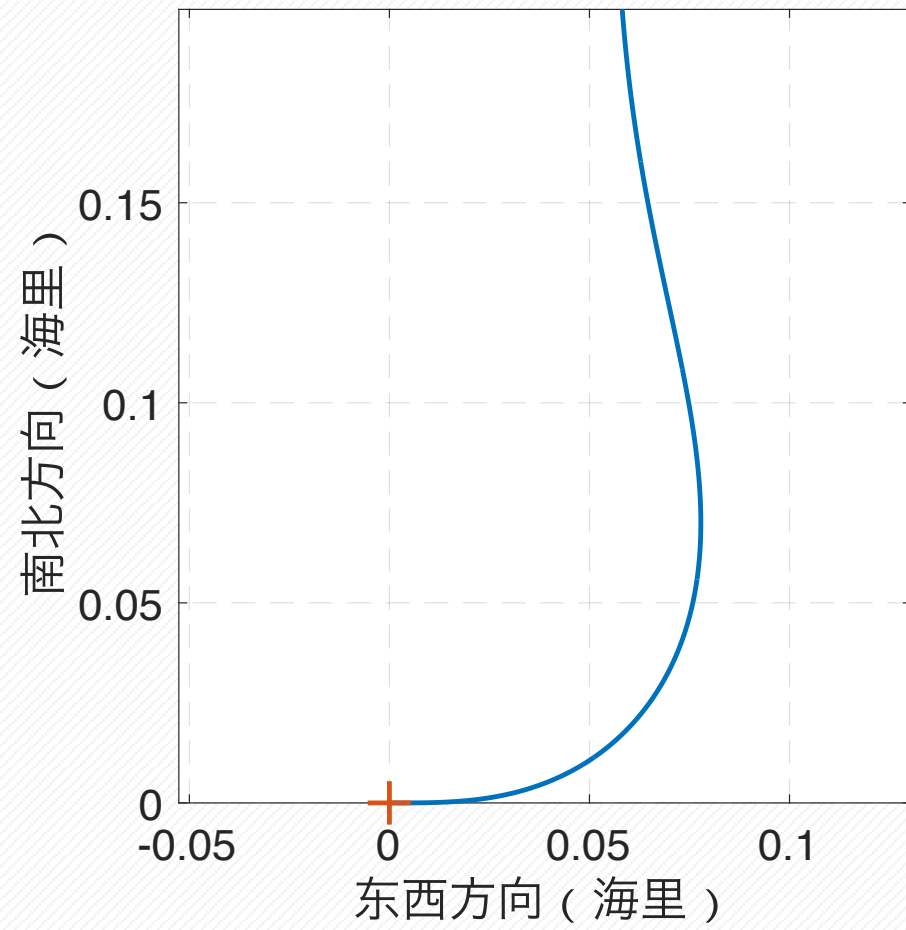
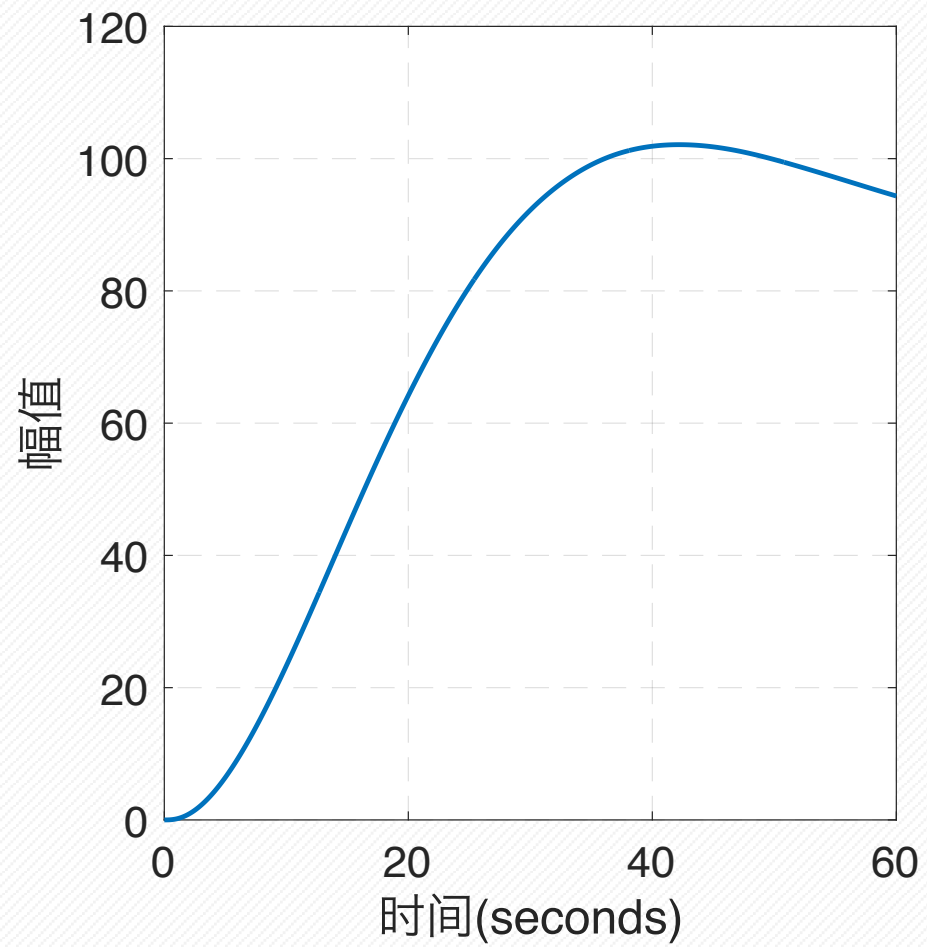
直角转向性能 ($K=10$, $V=15$ 节)



直角转向性能 ($K=5$, $V=15$ 节)



直角转向性能 ($K=1$, $V=15$ 节)



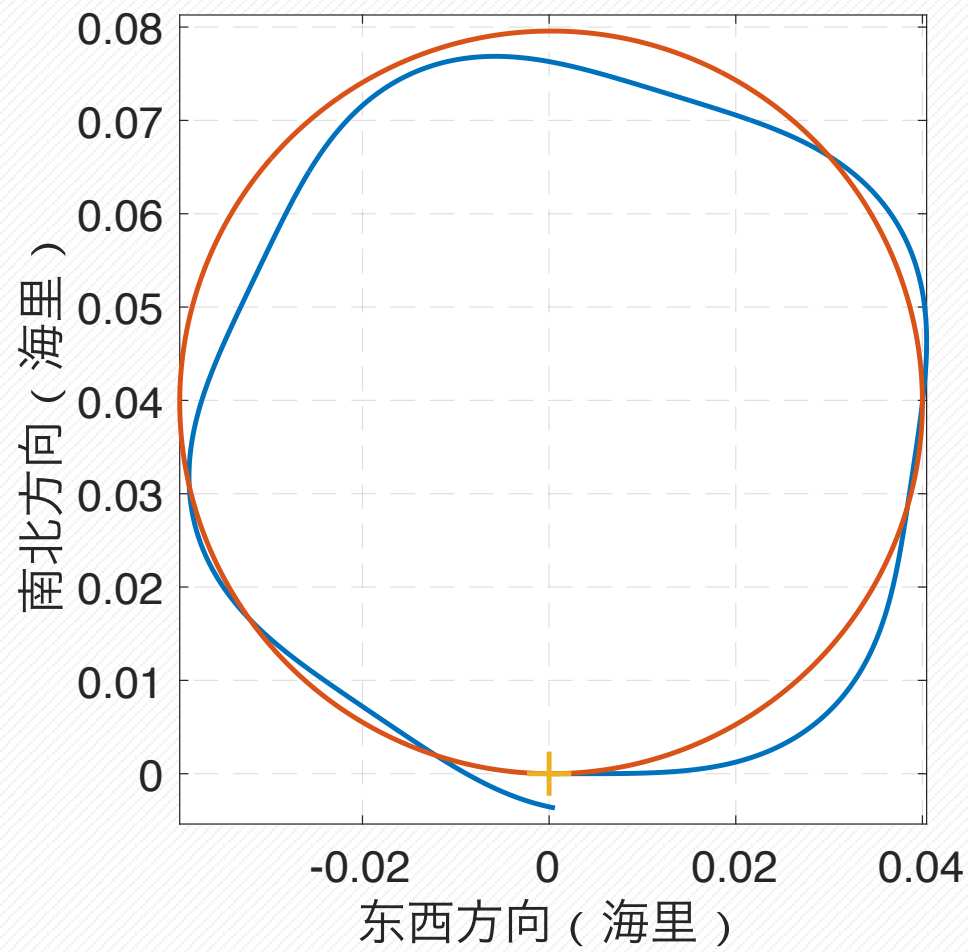
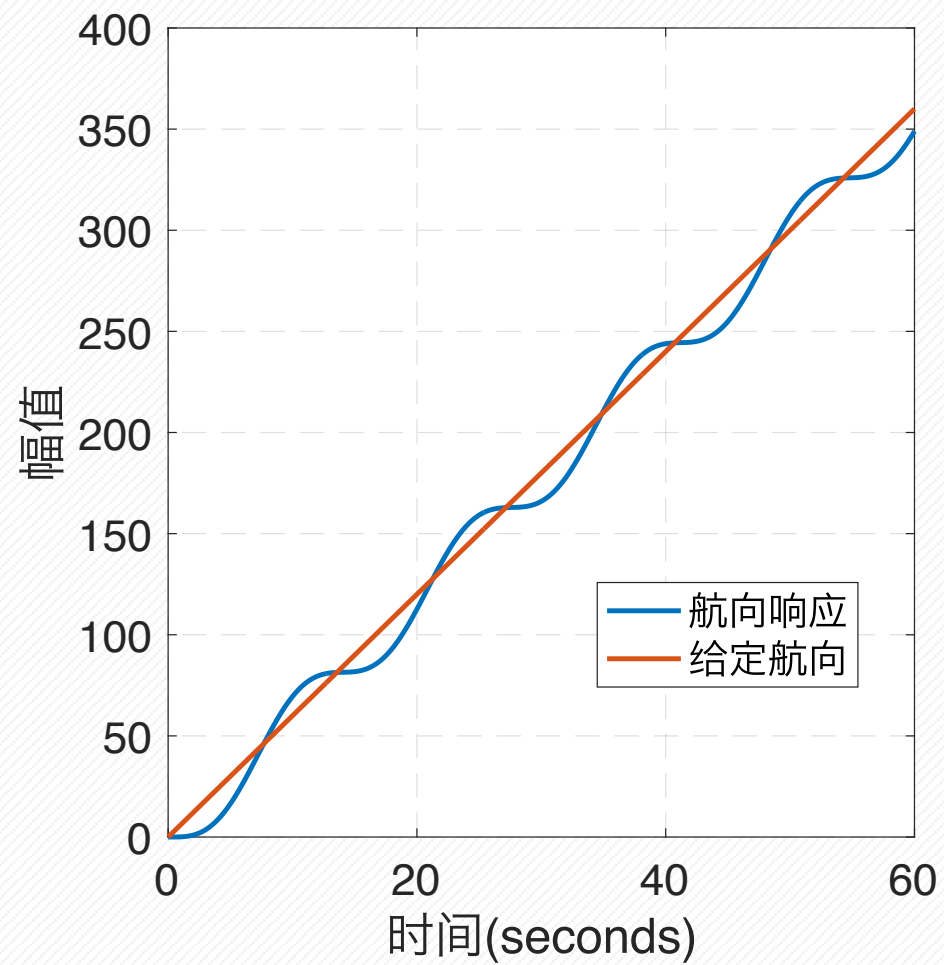
直角转向性能

代码

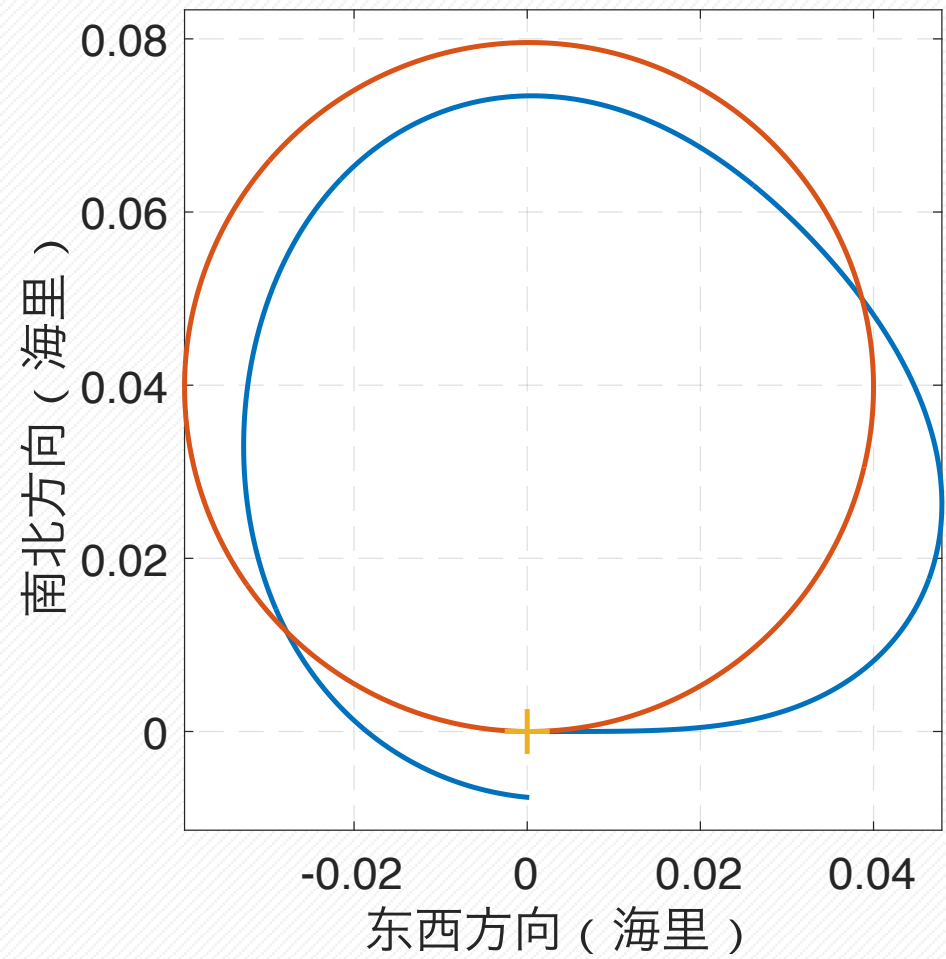
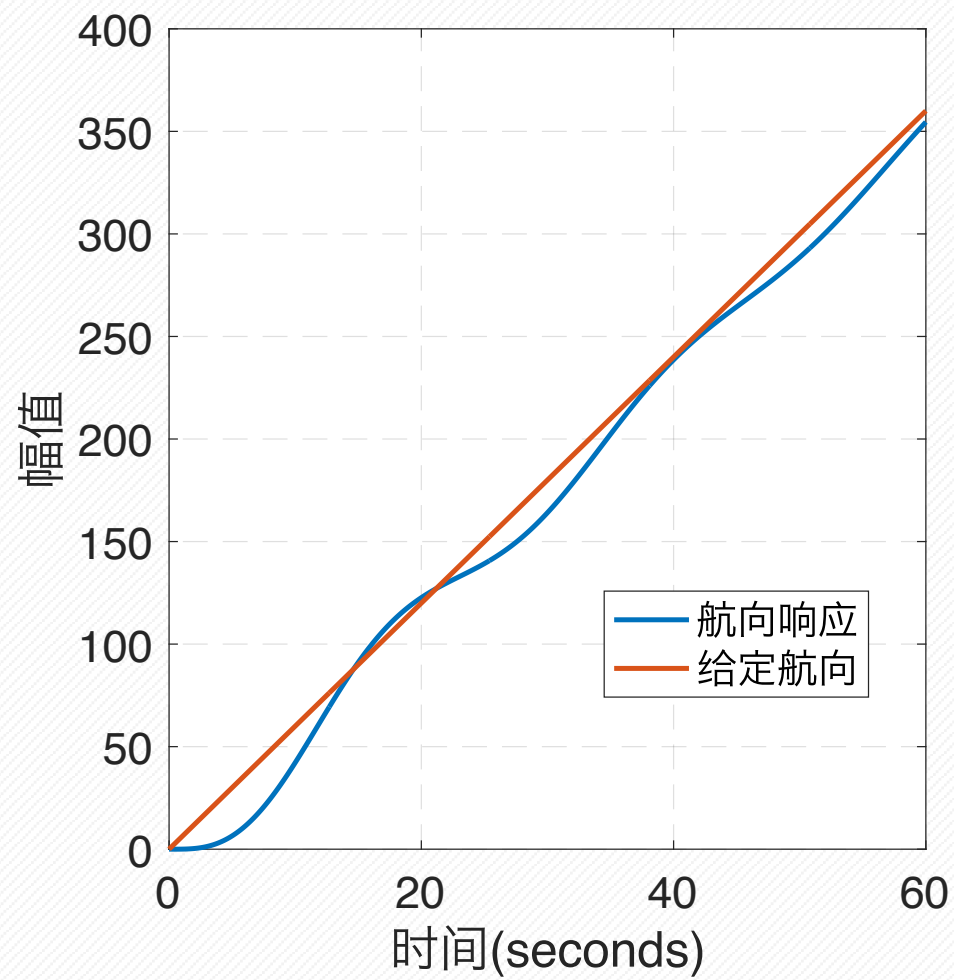
```
K = 1;
V = 15/3600; % Knots
dt = 0.1; % Time interval
G1 = 3.43/(s+2.14375);
G2 = 0.005/(s*(s+0.1));
PhiR = K*G1*G2/(1+K*G1*G2);
PhiN = G2/(1+K*G1*G2);
t = 0:dt:60;
u = 90*ones(size(t));
[y1,t] = lsim(PhiR,u,t);
u = sin(2*pi/10.*t+pi/6);
[y2,t] = lsim(PhiN,u,t);
y = y1 + y2;
tiledlayout(1,2);
fig = gcf;
fig.Position = [75,349,892,420];
nexttile
plot(t,y1+y2);
```

```
xlabel('时间(seconds)')
ylabel('幅值');
grid on
ax = gca;
ax.GridLineStyle = "--";
nexttile
x0 = [0,0];
xd = [V*cosd(y),V*sind(y)]*dt;
x = x0 + cumsum(xd);
plot(x(:,1),x(:,2));
hold on
plot(x0(1),x0(2),'Marker','+', 'MarkerSize',18);
xlabel('东西方向 (海里)')
ylabel('南北方向 (海里)');
axis equal
grid on
ax = gca;
ax.GridLineStyle = "--";
ax.GridLineWidth = 2; % R2023a
```

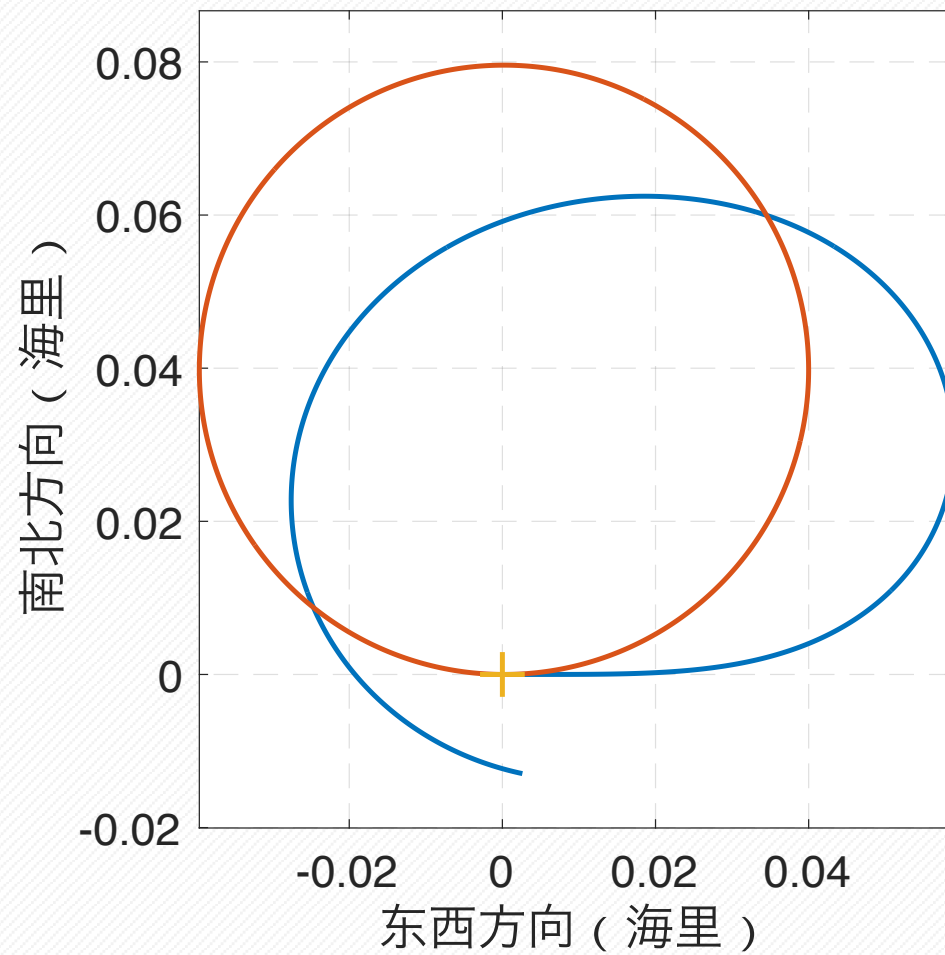
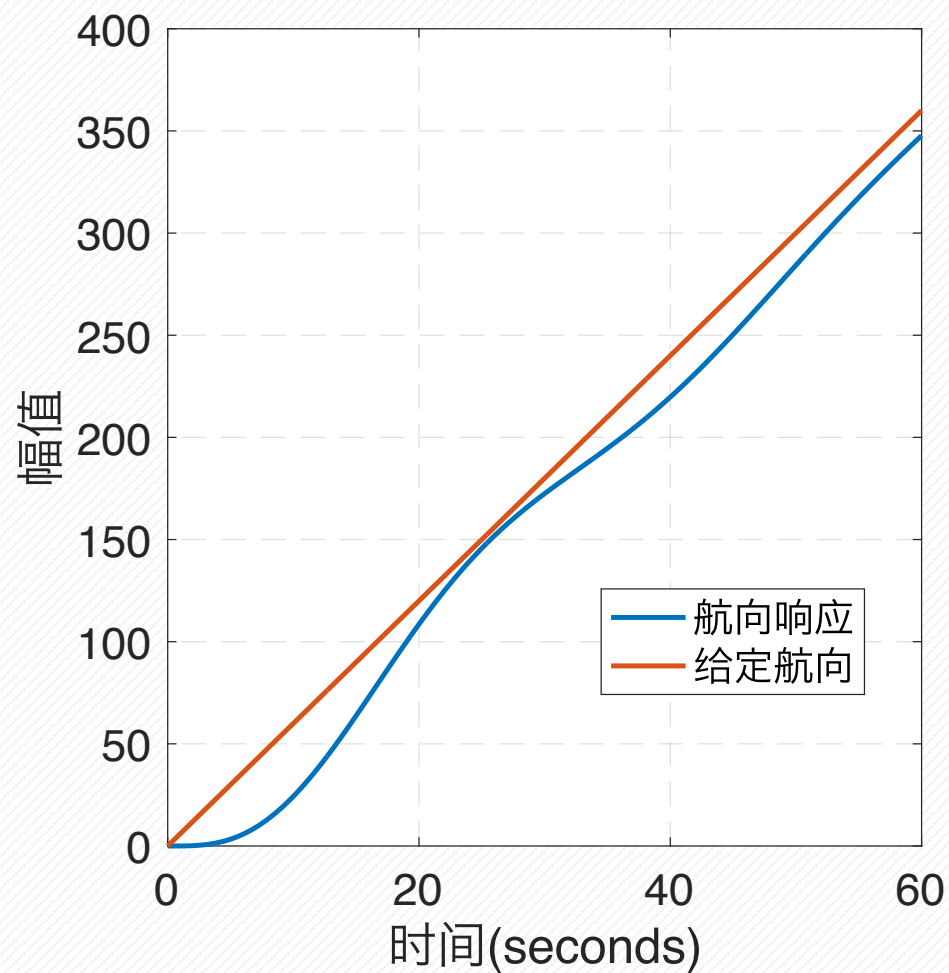
回转性能 ($K=28$, $V=15$ 节)



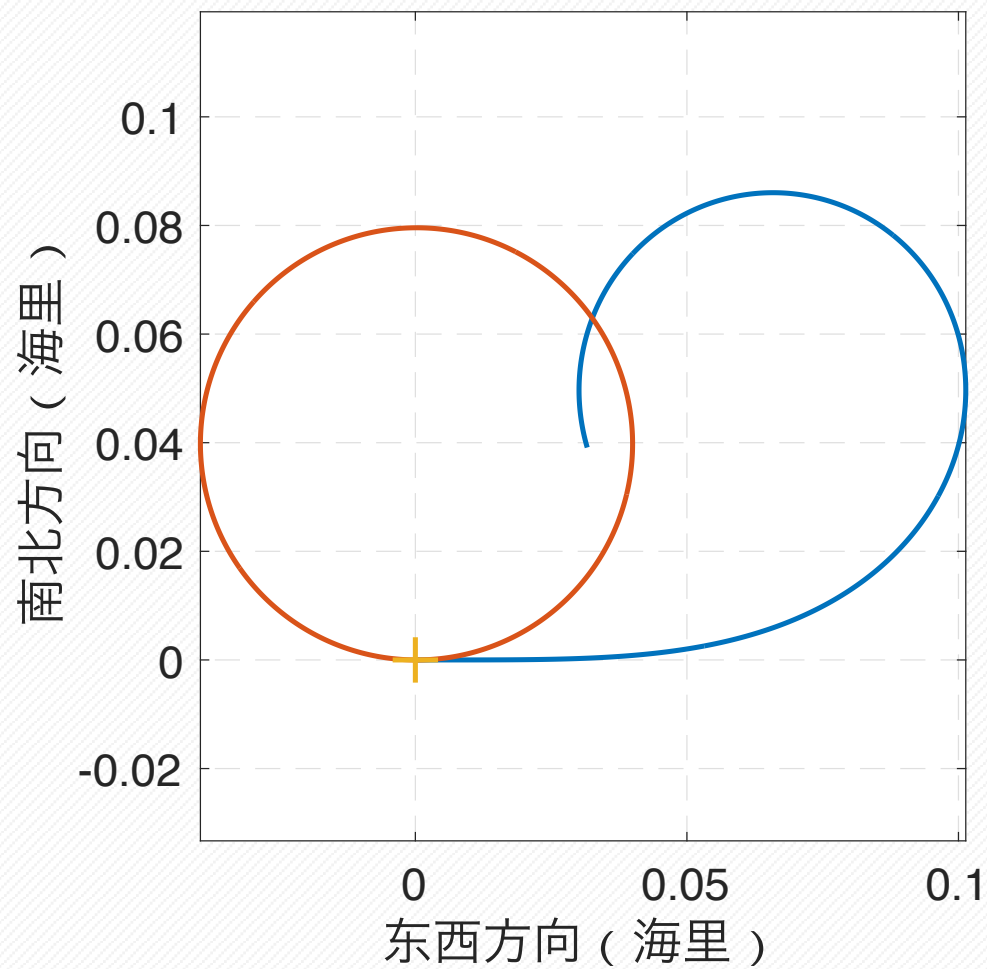
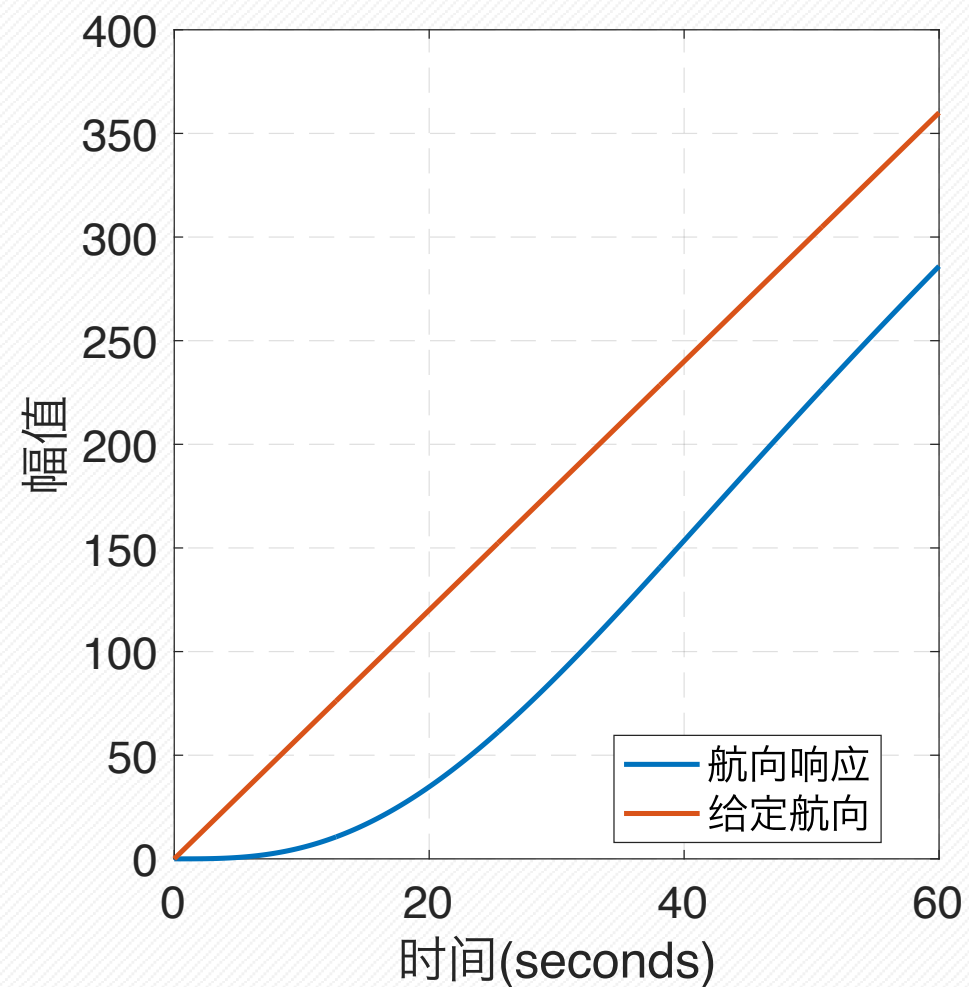
回转性能 ($K=10$, $V=15$ 节)



回转性能 ($K=5$, $V=15$ 节)



回转性能 ($K=1$, $V=15$ 节)



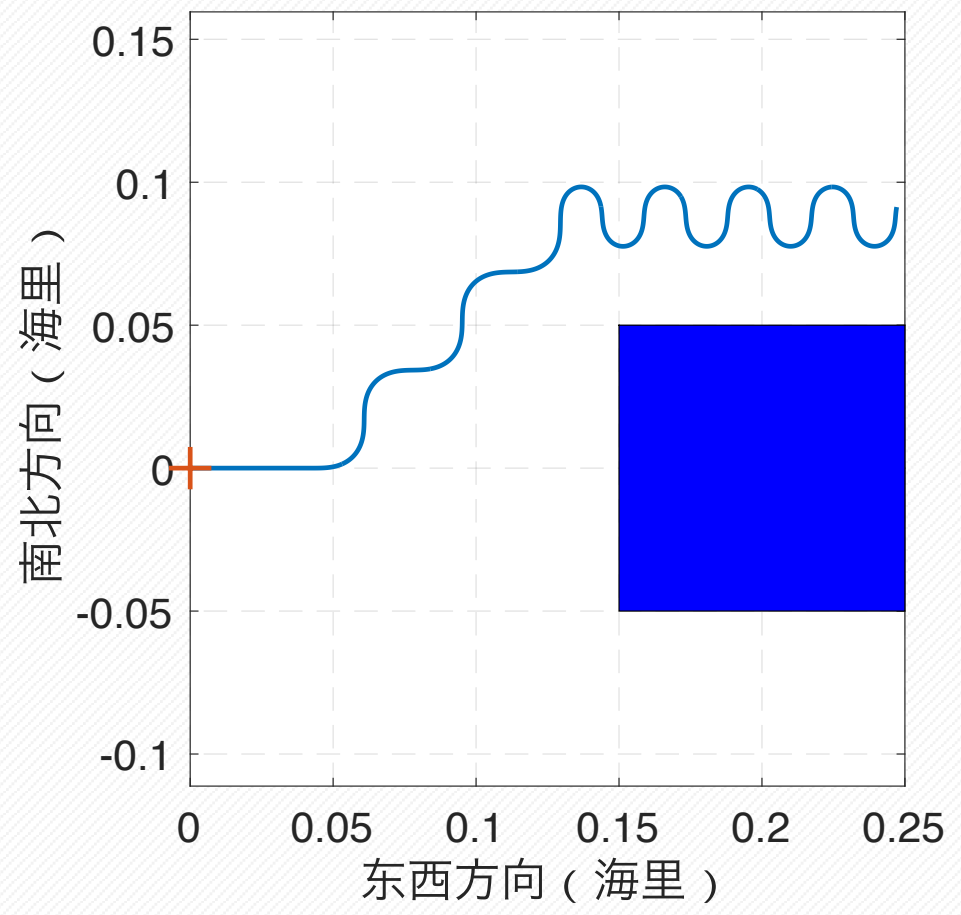
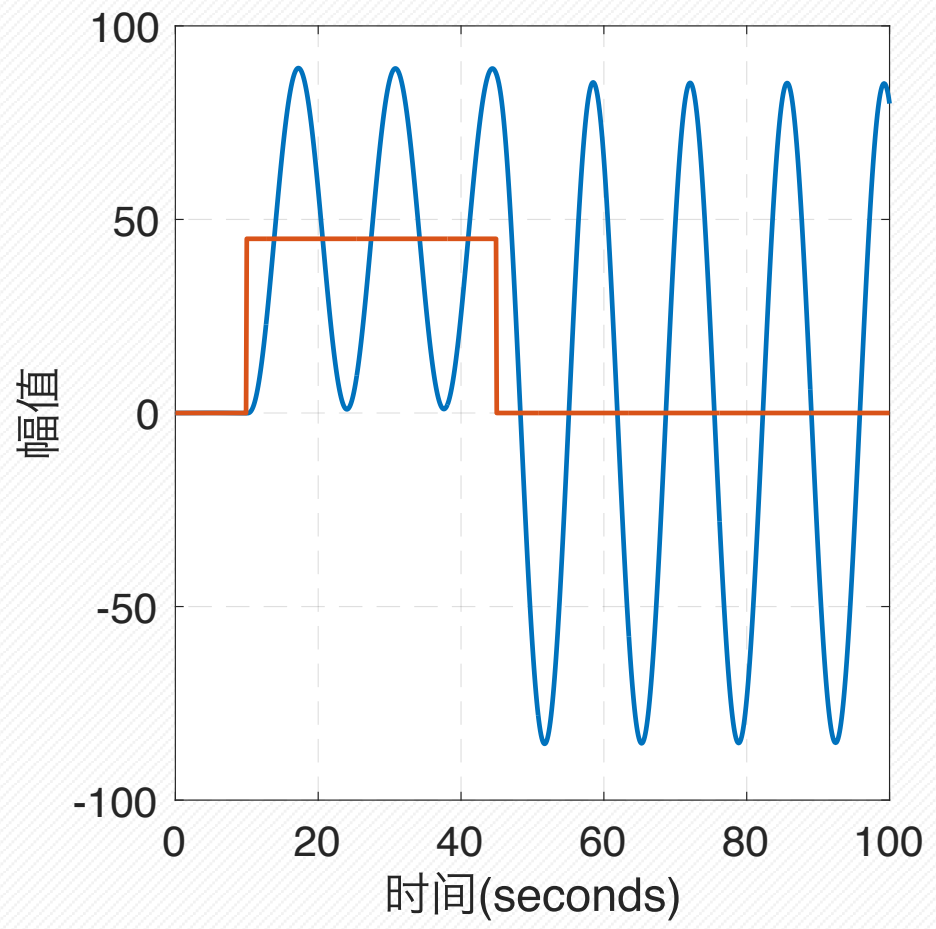
回转性能

代码

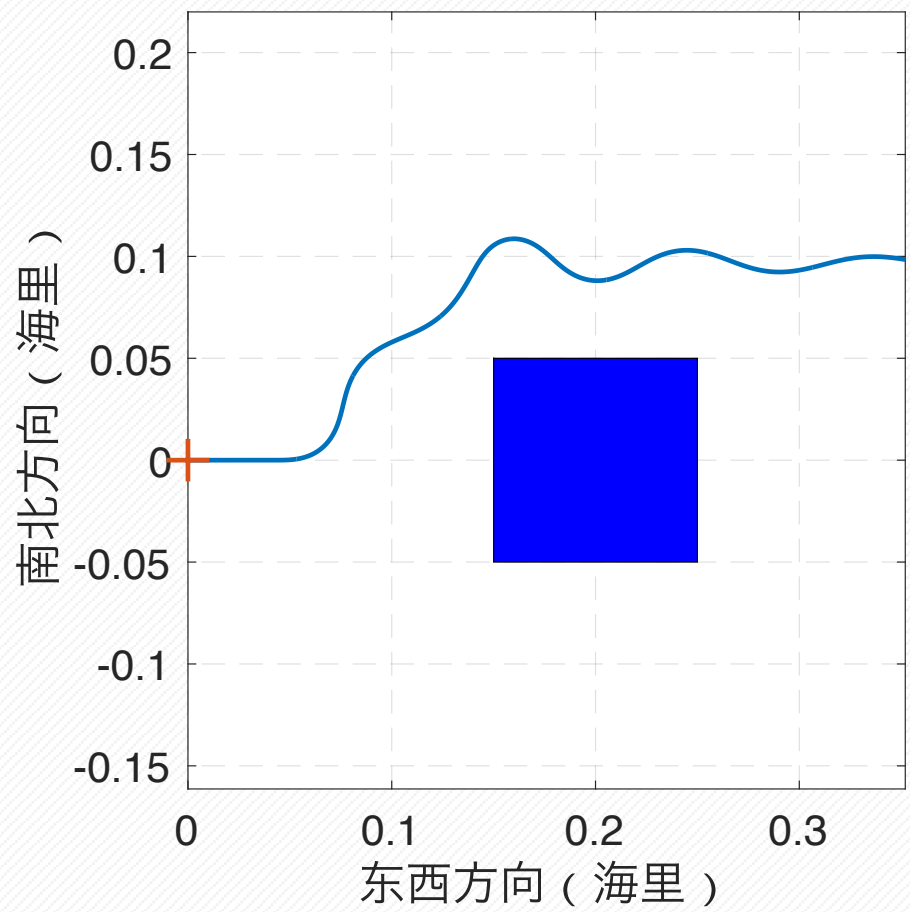
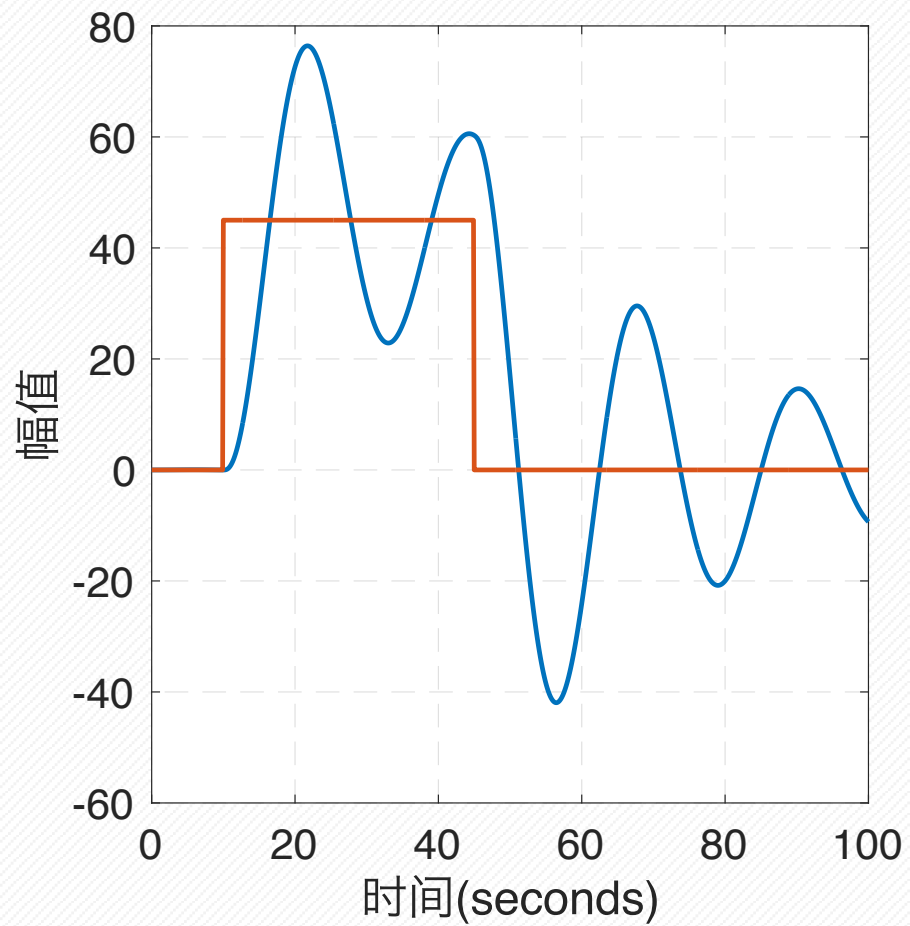
```
close all
V = 15/3600; % Knots
K = 28;
G1 = 3.43/(s+2.14375);
G2 = 0.005/(s*(s+0.1));
Phi = feedback(K*G1*G2,1);
dt = 0.1;
t = 0:dt:60;
u = t'*6;
[y,t] = lsim(Phi,u,t);
x0 = [0,0];
tiledlayout(1,2);
fig = gcf;
fig.Position = [75,349,892,420];
nexttile
plot(t,[y,u]);
xlabel('时间(seconds)')
ylabel('幅值');
```

```
grid on
ax = gca;
ax.GridLineStyle = "--";
legend({'航向响应','给定航向'}, 'Location','best');
nexttile
xd = [V*cosd(y),V*sind(y)]*dt;
x = x0 + cumsum(xd);
plot(x(:,1),x(:,2));
hold on
xd = [V*cosd(u),V*sind(u)]*dt;
x = x0 + cumsum(xd);
plot(x(:,1),x(:,2));
plot(x0(1),x0(2),'Marker','+','MarkerSize',18);
% patch([1,1.1,1.1,1]-
0.85,[0.05,0.05,-0.05,-
0.05],'b');
xlabel('东西方向 (海里)')
ylabel('南北方向 (海里)');
axis equal
grid on
ax = gca;
ax.GridLineStyle = "--";
```

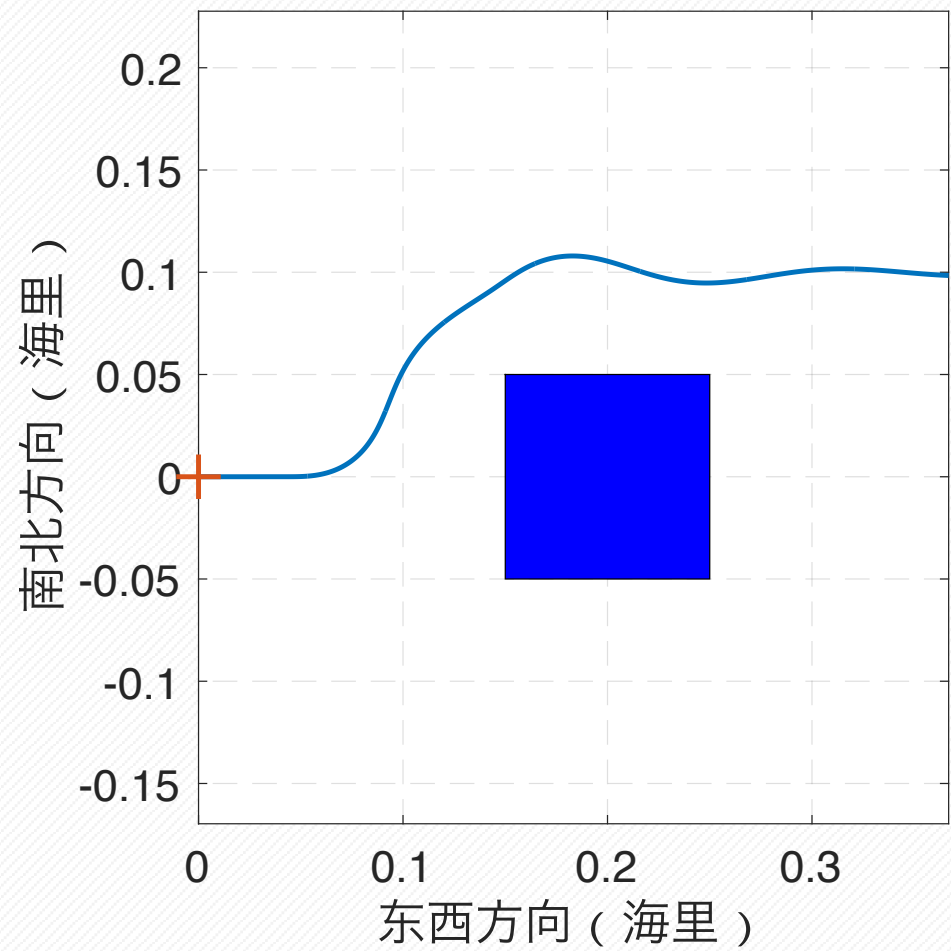
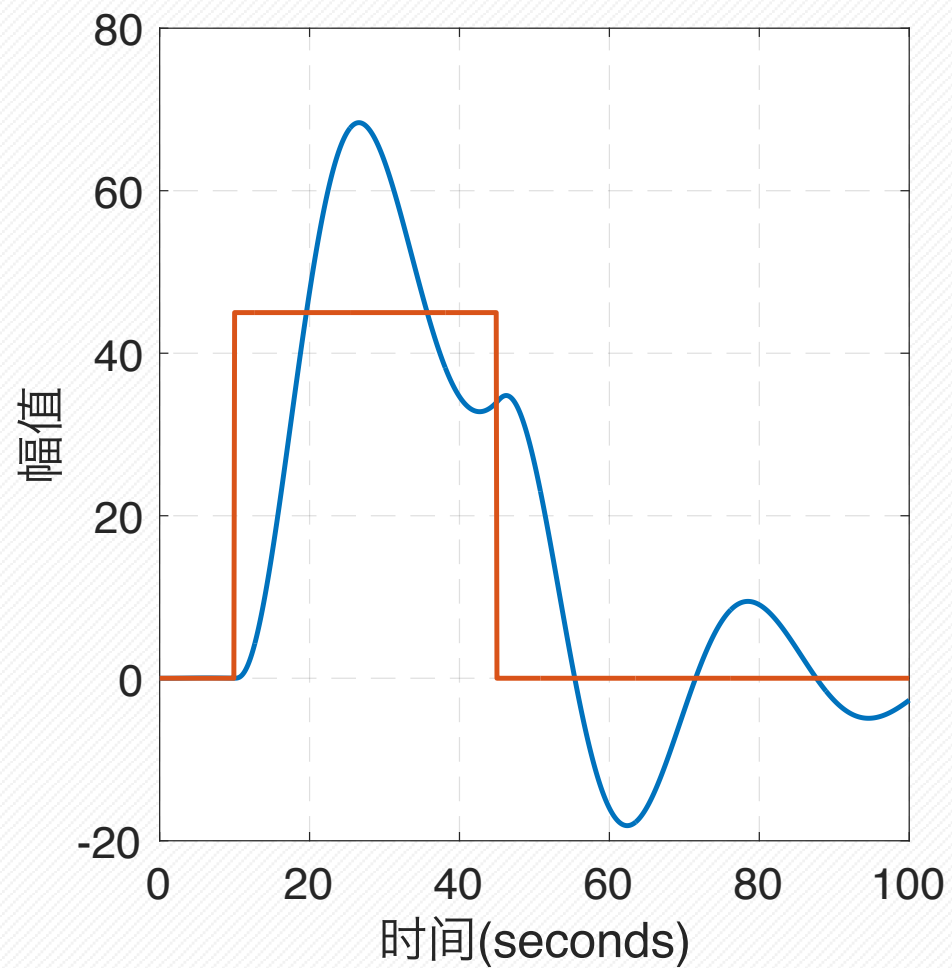
避障性能 ($K=28$, $V=15$ 节)



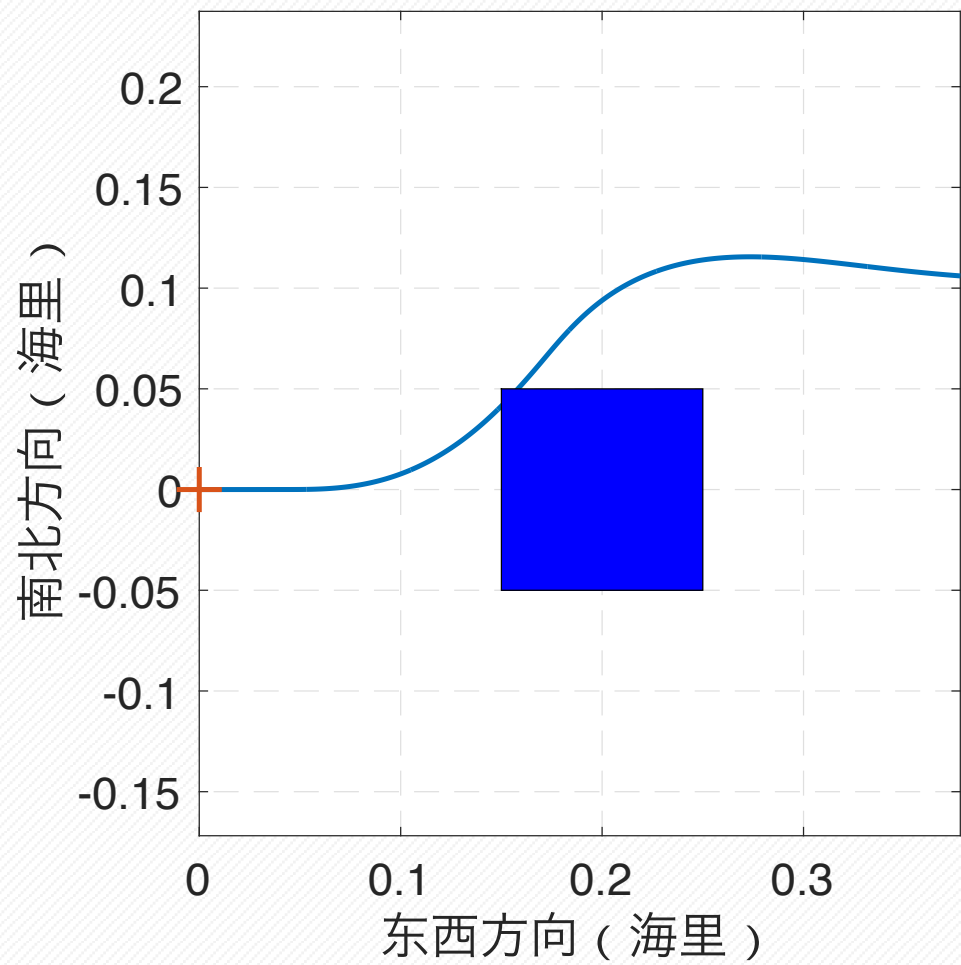
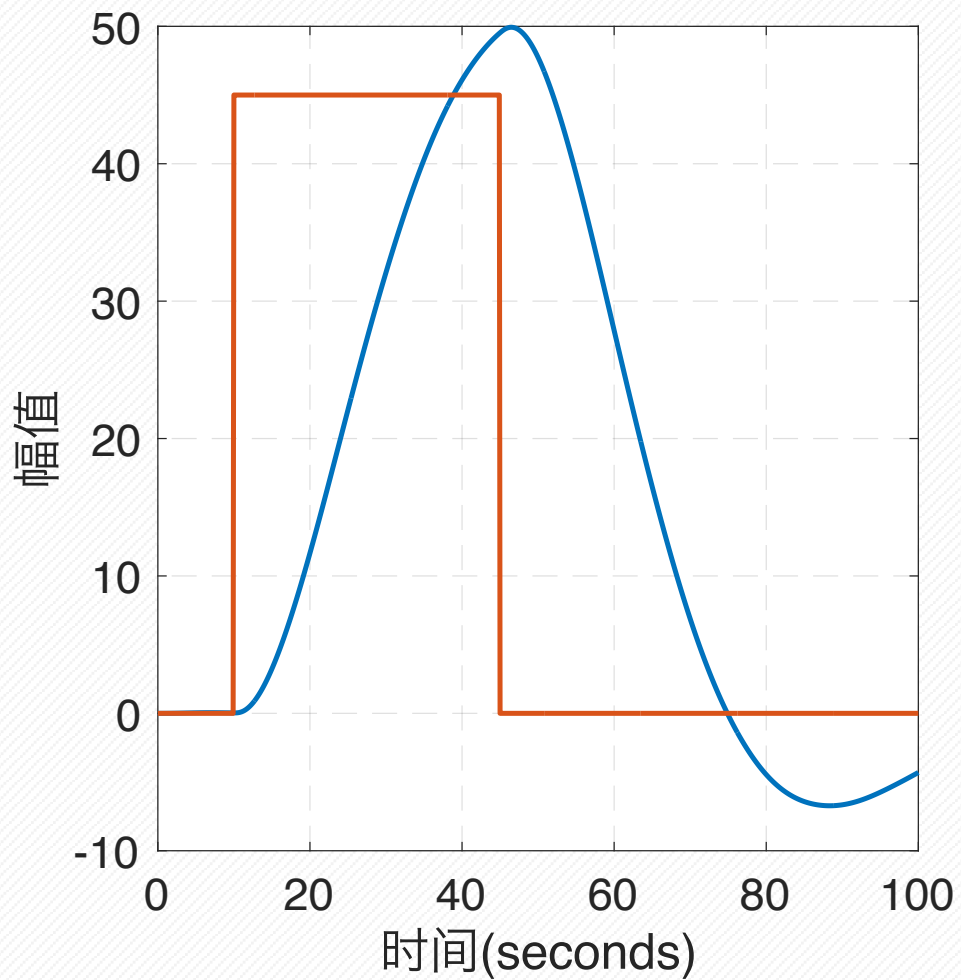
避障性能 ($K=10$, $V=15$ 节)



避障性能 ($K=5$, $V=15$ 节)



避障性能 ($K=1$, $V=15$ 节)



避障性能

代码

```
K = 1;
V = 15/3600; % Knots
dt = 0.1; % Time interval
G1 = 3.43/(s+2.14375);
G2 = 0.005/(s*(s+0.1));
PhiR = K*G1*G2/(1+K*G1*G2);
PhiN = G2/(1+K*G1*G2);
t = 0:dt:100;
u1 = zeros(size(t))';
u1(101:450) = 45;
% u1(301:end) = -45;
[y1,t] = lsim(PhiR,u1,t);
u2 = sin(2*pi/10.*t+pi/6);
[y2,t] = lsim(PhiN,u2,t);
y = y1 + y2;
tiledlayout(1,2);
fig = gcf;
fig.Position = [75,349,892,420];
```

```
nexttile
plot(t,[y,u1]);
xlabel('时间(seconds)');
ylabel('幅值');
grid on
ax = gca;
ax.GridLineStyle = "--";
nexttile
x0 = [0,0];
xd = [V*cosd(y),V*sind(y)]*dt;
x = x0 + cumsum(xd);
plot(x(:,1),x(:,2));
hold on
plot(x0(1),x0(2),'Marker','+','MarkerSize',18);
patch([1,1.1,1.1,1]-
0.85,[0.05,0.05,-0.05,-
0.05],'b');
xlabel('东西方向 (海里)');
ylabel('南北方向 (海里)');
axis equal
grid on
ax = gca;
ax.GridLineStyle = "--";
```